Worker Responses to Immigration Across Firms: Evidence From Colombia

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Abstract

This paper examines the short-run labor market effects of the mass arrival of Venezuelan migrants into Colombia using matched employer-employee data. After immigrants arrive, formal employment declines among minimum-wage natives, while those earning above the minimum face negative wage effects. Small, lower-paying firms drive these effects, consistent with a model in which they can hire workers informally. Using machine learning, I show that firm-level factors are important predictors of heterogeneity in worker-level impacts. These findings highlight how informality can amplify the role of firms in shaping immigration effects.

Keywords: Immigration, Minimum wages, Formal labor markets, Causal forest.

JEL Codes: F22, O15, O17, R23.

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1 Introduction

Over the past decade, several countries around the globe have experienced substantial population outflows, notably Afghanistan, Ukraine, Syria, and Venezuela. The majority of these migrants and refugees are moving to neighboring, developing countries. As of June 2023, 75% of the 110 million forcibly displaced individuals were hosted in low- and middle-income countries (UNHCR, 2023). The labor market in these contexts features the coexistence of formal and informal labor in production (Ulyssea, 2018) and the prevalence of small businesses (McKenzie, 2017). As migrants tend to disproportionately work informally and in small firms, immigration effects on formal natives can be shaped by the firm they work for, beyond the traditional skill-substitution channel (Borjas, 1999). In fact, there is a growing literature that documents the relevant role of firms when analyzing immigration adjustments in various developed countries (Amior and Stuhler, 2022; Beerli et al., 2021; Clemens and Lewis, 2022; Orefice and Peri, 2024). This paper shows that, in settings with high informality and binding minimum wages, firms play a more central role in determining how formal workers and the formal labor market respond to immigration shocks.

To do so, I examine the labor market impacts of one of the most significant episodes of immigration in recent history: the Venezuelan mass migration to Colombia, which brought more than 2 million migrants with skills similar to those of natives and who had access to work permits through massive regularization programs, although the vast majority end up working informally. By exploiting the uneven arrival of migrants across local labor markets and tracking workers over time, I quantify worker-level impacts and study heterogeneity across firm-level characteristics.[1] This paper differs from previous work (Delgado-Prieto, 2024), which uses survey data from the same setting. Rather than re-estimating the formal–informal sector regional responses documented there, I examine how the worker-level effects of immigration are distributed across firms, leveraging the universal coverage of formal workers and firms I have access to and complementing the analysis

[1] I address these compositional changes in the employed population more carefully with this data, which are typically omitted when using regional outcomes. For instance, recent papers emphasize that when a specific group of workers leaves the labor force or relocates to other regions, wage estimates are not accurately identified (Borjas and Edo, 2026; Dustmann et al., 2026).

with census counts that accurately measure the stock of migrants in each city.[2] To address endogenous sorting of migrants into economically favorable areas, I exploit two distinct instruments: past Venezuelan settlement locations and the local distance to the nearest crossing bridge between Colombia and Venezuela. Using these instruments in a difference-in-differences research design (DiD-IV), I document a negative impact on individual formal employment and native wages in the short to medium term.

The negative employment effects are driven by workers earning the minimum wage, the least educated within the formal sector. For these low-wage workers, a 1 percentage point (pp) increase in the immigration shock in a given labor market decreases the probability of formal sector employment by 1.5 pp. These workers are likely to transition towards informality, but may also move into unemployment or inactivity. In this context, the relatively high and binding minimum wage limits the space for downward wage adjustments and increases the risk of job displacement. Approximately 40% of all formal workers, including self-employed individuals, earned the minimum wage in 2015, and in small firms this share exceeds 50%. For wages of workers employed in both 2015 and the comparison year, the negative impact, in contrast, primarily affects those earning above the minimum wage.[3] Notably, these effects are concentrated in workers employed in small formal firms, which face binding minimum-wage constraints and hire a larger share of their workers informally.

To rationalize this, I construct a model of an imperfectly competitive labor market based on Card et al. (2018) that features heterogeneous firms, some of which pay a minimum wage. The model has two goals: to show why an aggregate informal labor shock passes through directly to formal employment and wages within formal firms, and to derive how that pass-through varies by firm, which motivates the firm-level focus of the empirical analysis. This model incorporates labor input costs similarly to Ulyssea (2018), while allowing for imperfect substitution of labor inputs, as in Delgado-Prieto (2024). The model indicates that aggregate substitutability between formal and

[2]Throughout this paper, formal workers refer to workers who contribute to the health system in Colombia.

[3]Wage estimates are less robust than the employment estimates and lose significance once border areas are excluded from the sample.

informal workers in production must be high to observe negative responses in formal employment and wages.[4] It also reveals that firm-level responses to immigration depend on their reliance on informal labor for production, ultimately suggesting that smaller firms have more elastic formal labor demand than larger firms. Consistent with the model, I find with the administrative data that workers in smaller firms experience more adverse employment impacts than those in larger firms, and with complementary survey data, I find higher informal shares in smaller firms after migrants arrive.

I then estimate the canonical "AKM" regression (Abowd et al., 1999) to construct firm fixed effects (FEs) representing firm pay premiums, and worker FEs as a proxy for workers' constant skills portable across employers. I find that native workers in lower-paying firms are more adversely affected in terms of employment than those in high-paying firms. Regarding the wage results, a potential explanation for the observed negative effects is the reallocation of workers across firms, whereby workers may move from high-paying to low-paying firms, or vice versa (Gyetvay and Keita, 2023). But I find no differential sorting of native workers after the immigration shock, potentially due to institutional factors or sluggish economic conditions. There is also no evidence that formal workers are relocating to other regions following the arrival of immigrants. Last, I examine complementary firm-level outcomes that highlight other margins of adjustment. I find that firms recruit a smaller share of their hires from other formal firms, especially those that, at baseline, recruit the least from them. Additionally, areas with higher migrant inflows experience higher formal firm exits, while formal firm entry remains unchanged.

To discover which variables predict these differences, I systematically estimate heterogeneous immigration effects using machine learning methods (Athey and Imbens, 2016; Athey et al., 2019). I implement causal forests to obtain reduced-form estimates from random subsamples, thereby identifying the worker and firm variables that are most important for predicting the heterogeneity in worker-level effects. Using this algorithm, I first identify the subgroups most affected by immigration, both in employment and wages. Then, based on the frequency of these variables across all

[4]This follows the first Hicks-Marshall rule of Derived Demand: "The demand for anything is likely to be more elastic, the more readily substitutes for the thing can be obtained" (Hicks, 1932).

decision trees in the causal forests, I compute a variable importance statistic. I show that, with this measure, firm pay premiums and firm size are among the variables the algorithm splits on most often, an indication that firm characteristics help predict the heterogeneity in employment and wage effects alongside worker characteristics, such as job tenure, age, sex, or wages in the pre-shock period. All in all, both regression analysis and causal forests complement the theoretical framework's prediction that workers' adjustments to immigration shocks depend on how firms respond to these shocks, and that response might be influenced by the institutional factors in which they operate, for instance, through binding minimum wage constraints or a looser regulatory environment that fosters informality.

These findings reflect only short- and medium-term effects from 2015 to 2019. As workers and firms adapt over time, a longer-term analysis may demonstrate quantitatively different results (Monras, 2020). However, such analysis would be constrained in this context due to the onset of the COVID-19 pandemic. Nevertheless, the findings indicate the need for policies to support native workers. Given Colombia's relatively high minimum wage, reducing other labor costs for formal employers could help sustain formal employment (Morales and Medina, 2017; Kugler et al., 2017). Additionally, stricter enforcement of fines for hiring informally could deter its use (de la Parra and Bujanda, 2024).

Related Literature. This paper contributes to different strands of the labor economics literature. First, this paper contributes to the growing literature on the impact of international migration in developing countries. It joins a small set of studies using matched employer-employee or firm-level data to study immigration shocks outside high-income countries (Altındağ et al., 2020; Gulek, 2024; Bandiera et al., 2023; Osuna Gomez and Medina-Cortina, 2026; Rozo and Winkler, 2021), and is, to my knowledge, the first to do so for the Venezuelan immigration in Colombia.[5] I document negative wage and employment effects for formal natives at the worker level, whereas evidence from related studies on Venezuelan immigration comes from aggregate patterns from

[5] Altındağ et al. (2020) and Gulek (2024) study Syrian refugees and firms in Turkey, Bandiera et al. (2023) and Osuna Gomez and Medina-Cortina (2026) study deportation-induced labor supply shocks in El Salvador and Mexico, and Rozo and Winkler (2021) studies internally displaced in Colombia.

surveys and is mixed. In Colombia, Bonilla-Mejía et al. (2024) and Delgado-Prieto (2024) report negative regional employment effects in the formal sector, while Caruso et al. (2021) and Lebow (2022) find only imprecise negative effects on regional formal wages.[6] By contrast, studies in Turkey document positive regional employment effects for men in the formal sector (Del Carpio and Wagner, 2015; Ceritoglu et al., 2017; Aksu et al., 2022), and evidence from Peru indicates positive effects for highly educated natives (Groeger et al., 2024). Beyond differences in institutional factors and migrant characteristics, these studies rely on cross-sectional surveys for their regional-level analysis. Although labor force surveys provide crucial information about the informal sector, they are typically restricted to broad geographic or firm-size categories, making it challenging to test firm-level mechanisms. Moreover, worker and firm characteristics are interrelated through sorting (e.g., minimum-wage workers often work in smaller firms), so the heterogeneity in surveys might be correlated with worker composition. With the administrative data, I can control for this directly. This also addresses the conceptual differences between regional and individual labor-market responses emphasized by Dustmann et al. (2026), which helps explain why my findings are novel relative to prior studies on Colombia, Peru, and Turkey.

Second, it contributes to the literature analyzing how firms shape native and migrant outcomes. Several papers emphasize that firms influence workers' outcomes through different channels. For example, Beerli et al. (2021) examines the abolition of immigration restrictions and shows that this led to the growth of Swiss firms, thereby improving the wages and job opportunities of highly educated natives. Studies using the H-1B visa lottery also find direct effects on native workers (Doran et al., 2022; Kerr and Lincoln, 2010; Peri et al., 2015; Mayda et al., 2018).[7] For the low-skill H-2B visa lottery, in contrast, Clemens and Lewis (2022) finds that firms able to hire low-skill immigrants increase production with zero employment effects on natives. Additionally,

[6]Other related Colombian studies include Santamaria (2020), which finds no wage or employment effects on native workers, Peñaloza Pacheco (2019), which documents wage declines concentrated among low-skilled workers near the border, and Lebow (2024), which shows substantial occupational downgrading among Venezuelan migrants.

[7]Doran et al. (2022) shows that winning firms crowd out existing workers with H-1B visa holders, while Kerr and Lincoln (2010) find that expansions raise skilled immigrant employment without displacing natives. Peri et al. (2015) document positive native wage and employment effects in STEM-intensive cities, and Mayda et al. (2018) show that quota effects operate mainly through migrant selection rather than displacement.

there is evidence that firms attract natives from lower- to higher-paying firms post-immigration in developed countries (Orefice and Peri, 2024; Gyetvay and Keita, 2023).[8] In this respect, my findings show that immigration effects are concentrated on natives working in small and low-paying firms, similar to Amior and Stuhler (2022), though the mechanisms differ. In my paper, this implication stems from the highly informal context with binding minimum wages, where firms jointly hire formal and informal workers for production. In contrast, Amior and Stuhler (2022) argues that small, low-paying firms in Germany tend to hire the most migrants because of lower reservation wages, thereby crowding out natives as they exert monopsony power and move along their labor demand curve. Together, these two novel mechanisms indicate which firms respond more to immigration shocks and how they shape aggregate responses.

Third, I contribute to the literature that estimates the individual impacts of immigration. Early studies, such as Bratsberg and Raaum (2012) and Foged and Peri (2016), exploited licensing requirements in the Norwegian construction sector and refugee dispersal policies in Denmark, respectively, to estimate worker-level or individual effects of immigration.[9] By incorporating all movements of natives between areas and excluding employment inflows, my analysis reduces the attenuation of wage estimates highlighted by Borjas (2006). This allows me to understand the main drivers of labor market adjustments to immigration, while implementing a machine-learning method that estimates heterogeneous effects in a data-driven manner.

Lastly, I complement the literature on how workers respond to other labor demand shocks, such as those coming from import competition (Autor et al., 2014), local unemployment shocks (Yagan, 2019; Redondo, 2022), or mass layoffs (Gulyas and Pytka, 2026; Yakymovych et al., 2022). My findings show that formal firms in highly informal contexts exert stronger wage and employment effects on formal natives, both theoretically and empirically, aligning with Gulyas and Pytka (2026), which uses causal forests and finds higher earnings losses among workers in

[8]Regarding migrant outcomes, Arellano-Bover and San (2026) and Dostie et al. (2021) find that firm pay premiums explain around one-fifth of the immigrant-native wage gap in Israel and Canada, respectively. Moreover, Marchal et al. (2023) finds that these gaps narrow depending on the exporting activity of the employing firm, as migrants have an informational rent when the exporting destination coincides with their region of origin.

[9]More recent papers that quantify worker-level effects of immigration include Hoen (2020) for Norway, Ortega and Verdugo (2022) for France, and Kuosmanen and Meriläinen (2022) for Finland.

high-paying firms post-displacement.[10]

The remainder of the paper is structured as follows. Section 2 discusses the Venezuelan crisis-induced immigration shock and describes the main datasets. Section 3 develops the theoretical framework. Section 4 outlines the empirical strategy and identification assumptions. Section 5 presents results at the worker level, broken down by worker characteristics. Section 6 shows results by firm characteristics. Section 7 introduces the machine learning approach and discusses the main findings. Finally, Section 8 concludes.

2 Institutional Context and Data

2.1 The Venezuelan Mass Migration

Colombia and Venezuela share an extensive land border, historically characterized by a dynamic relationship marked by frequent economic and cultural interactions. People often moved back and forth between the two countries, but generally, Colombians settled in Venezuela. This trend intensified after 1950, fueled by Venezuela's oil boom and Colombia's internal conflict. The economic opportunities presented by Venezuela's oil industry attracted many Colombians to emigrate, seeking better livelihoods and prospects for their families. Recently, this trend reversed as Venezuela's socio-economic and political crisis deepened. While some emigration began in 2013, outflows remained relatively modest until 2015. The situation changed markedly from 2016 onward, when migration accelerated sharply and eventually peaked around 2021. As a result, several countries in Latin America are receiving vast numbers of migrants, especially Colombia, Perú, and Ecuador (UNHCR, 2019). Colombia has by far been the main destination, hosting more than 2 million Venezuelans, including around 1 million of working age, equivalent to 4.1% of the working-age population as of 2019 (DANE, 2019).

The Venezuelan exodus is unprecedented in Latin America's recent migration history and is

[10] Yakymovych et al. (2022) uses similar methods with Swedish administrative data to identify workers most vulnerable to job displacement, finding that older, less-educated, and manufacturing workers experience the largest earnings losses.

globally comparable to the Syrian and Ukrainian exoduses. Turkey has been the major host country for Syrian refugees, with various papers analyzing the labor market impacts of this immigration shock (e.g., Del Carpio and Wagner (2015); Ceritoglu et al. (2017); Aksu et al. (2022); Gulek (2024)). However, the Colombian context differs from the Turkish one. Venezuelans speak the same language as Colombians, and while this migration combines both economic migrants and forcibly displaced refugees, those arriving in Colombia are more likely to be forced migrants than those settling in other countries.[11] They also have access to work permits from the Colombian government's generous open-border policy. Since 2018, irregular Venezuelans in Colombia have been eligible for the Special Permit of Permanence (PEP, by its Spanish acronym), which enables them to work for at least 2 years, provides access to essential services, and facilitates their integration into Colombian society. In 2021, to overcome the limitations of PEP renewals, the government enacted the Temporary Protection Statute for Migrants (ETPV, by its acronym in Spanish), offering up to 10 years of regularization. However, as of 2019, about 90% of Venezuelan immigrants remain employed in the informal sector (see Online Appendix Figure F.1a). Limited formalization can be driven by migrants' lack of skills certification and by employers' low incentives to formalize the employment of informal migrants, given unchanged enforcement costs of informality during regularization (Delgado-Prieto, 2026).[12] Consequently, Venezuelans are disproportionately concentrated at the lower end of the wage distribution (Delgado-Prieto, 2024), reflecting significant occupational downgrading despite their comparable or higher education levels relative to Colombians during the 2016–2019 period (see Online Appendix Figure F.1b).

The informal sector, therefore, absorbs most of the immigration shock in Colombia. In line with a positive labor supply shock, I find a significant negative effect of 1.9% on native informal wages for a 1 percentage point increase in the migrant share (see Online Appendix Table C.1). However, this shock also has direct effects on the formal sector. Because firms, especially smaller

[11] Because forced and economic migrants differ in location choice, labor supply behavior, and willingness to accept occupational downgrading (Becker and Ferrara, 2019; Brell et al., 2020), the substitutability with natives here may not extend to settings shaped by more economic migration.

[12] In Colombia, informal employment can imply penalties up to 400 times the monthly minimum wage. However, labor inspections are infrequent, and when workers report these violations to the Ministry of Labor, disputes generally favor employees (de la Parra et al., 2024).

ones, hire both formal and informal labor, formal employment is directly affected by the immigration shock due to their production technology. Spillover responses between the informal and formal sectors have been observed in other contexts as well, including Brazil (Corbi et al., 2021), Indonesia (Kleemans and Magruder, 2018), and Turkey (Gulek, 2024). Moreover, certain formal firms or self-employed workers may face increased competition in product or service markets from informal firms with lower labor costs (Rozo and Winkler, 2021; Bandiera et al., 2023). Therefore, examining how formal workers adjust across the firm distribution is crucial for understanding who is most affected by immigration. I also investigate the effects of immigration using machine learning to identify the worker and firm characteristics that predict worker-level heterogeneity. The data's granularity enables a comprehensive evaluation of the interactions among immigration, minimum wages, and firms, and how these ultimately affect formal workers in developing countries.

2.2 Data

The main dataset I use in this paper is the *Planilla Integrada de Liquidación de Aportes* (PILA), which contains administrative records from the Colombian social security system managed by the Health Ministry (*Ministerio de Salud y Protección Social*). The PILA contains information on the universe of formal workers in tax-registered firms. It excludes informal workers and informal firms but includes self-employed formal workers. The PILA is based on workers' monthly contributions to the health system in Colombia, calculated according to their reported base income. Each observation is a worker-employer match for a given month in the year. The dataset contains worker-level information on labor income, sex, age, job type (employee or self-employed), foreign status, municipality, and the firm identifier for each job. In the main analysis, I focus on August of each year from 2012 to 2019.[13]

The other main dataset I use in this paper is the 2018 Colombian census (CNPV, by its acronym in Spanish). As Colombian authorities recognize the significance of the Venezuelan exodus, they

[13] I focus on August to ensure consistency with the 2018 census, whose data collection ended in October and therefore does not capture migrant arrivals thereafter, and to avoid seasonal distortions of other months in Colombia (e.g., December-January or March-April).

included migration questions in the most recent census, including the year of arrival for all migrants, which I use to construct the immigration shock. The census provides the most reliable source of information on the local stock of migrants in the country.

For the analysis, I constructed a dataset with all individuals recorded in PILA between 2012 and 2019, with rows representing individuals and columns their yearly variables. The dataset includes 18,430,987 workers who appeared in at least one of these eight years. I then restricted to full-time native workers aged 25 to 55 as of 2015, assigning the immigration shock to each worker based on their location in 2015. This restriction narrowed the sample down to 7,123,223 workers.[14]

I then transformed the municipality variable into a more standard definition of local labor markets by adjusting the methodology of Sanchez-Serra (2016).[15] The adjustment resulted in 109 functional urban areas (FUAs) after excluding small or rural municipalities, producing a sample of 6,706,035 workers.[16] This is the sample I use for the employment analysis over time (a balanced panel), as the worker can be employed or not in the comparison year. For the wage analysis, I further restricted the sample to workers with 30 days of employment per month to avoid mechanically lower wages and to those with positive wages, requiring that the worker be employed in the comparison post-treatment year. Thus, the sample varies slightly by year, making it an unbalanced panel. It is important to note that informal workers, who account for about half of all employed individuals in Colombia, are not observable in the administrative data.

Descriptive Statistics for Formal Workers. Online Appendix Table F.3 shows descriptive statistics for natives, foreigners, and Venezuelans with PEP permits broken down by age, sex, and wages

[14]Selecting workers observed in the base period excludes inflows of workers in the post-treatment period from the analysis. Also, part-time workers constituted less than 0.3% of the PILA workforce in 2015.

[15]The municipality variable in the PILA has the issue that some firms with establishments in smaller cities may report their workers under the nearest capital city where their main establishment is located. This could lead to a misrepresentation of local employment, but as long as it is unrelated to the local immigration shocks, it is less of a concern.

[16]The FUAs definition consists of the 53 most extensive urban areas in the country, defined from population grid data, municipal boundaries, inter-municipal commuting flows, as well as 56 municipalities with more than 2,500 formal workers according to the restricted 2015 sample. I exclude San Andrés, Cumaribo, Leticia, and San José del Guaviare from this definition as they belong to islands or the Amazon region. Online Appendix Table F.1 shows the sample distribution by FUAs, with only 5.9% of workers excluded from the analysis.

over time.[17] Venezuelans with PEP in the formal sector tend to be younger, predominantly male, and earn lower wages compared to both natives and other foreigners. The share of Venezuelans with PEP working in the formal sector remains small, suggesting that the impact of regularization on the Colombian labor market is limited (Bahar et al., 2021) due to a weak first-stage effect. Other foreigners, in contrast, earn substantially higher salaries than natives, yet this gap has been shrinking in recent years. The number of other foreigners in the formal sector has also risen, suggesting it may include Venezuelans holding other types of documentation.

Online Appendix Figure F.2 illustrates how binding the minimum wage is for a large share of formal workers, with around 40% of all formal workers earning it in 2015. [18] Importantly, the national minimum wage must increase, by law, by at least the inflation rate in the preceding year. This downward rigidity suggests why minimum-wage workers generally do not experience real wage declines during positive labor supply shocks or negative demand shocks, and why job displacement becomes more likely instead. During the analysis period (2015–2019), the minimum wage increased in real terms by less than 3% each year, mitigating concerns about additional employment impacts.[19]

3 Stylized Model

In this section, I develop a partial equilibrium model with heterogeneous firms and worker types to motivate how firms can shape immigration effects in a context of informality, and to guide the

[17] To identify foreign status in PILA, I use the type of document listed in the workers' health contribution. Workers with a national ID are classified as natives, those with the PEP documentation are classified as Venezuelan regularized migrants, and those with a foreign ID or passport are classified as foreigners. The PEP program was created in 2017, but PEP holders are identifiable in the PILA data used here only from 2018.

[18] To contribute to the social security system, workers must declare a labor income equal to or above the monthly minimum wage (up to 25 minimum wages), so many self-employed workers, who can decide what their base income is, report the minimum wage even if they earn more or less. Adjusting labor income is more rigid for employees, as it also affects the additional 20.5% of labor income that firms pay toward workers' health and pension contributions. If under-the-table payment agreements exist (Feinmann et al., 2022), any changes in these payments in response to the immigration shock might bias the observed wage effects.

[19] Online Appendix Table F.4 aggregates worker-level data into seven firm size categories, excluding self-employed workers, and reports descriptive statistics on sex, age, and wages by firm size. Online Appendix Figure F.3 presents a histogram of firms by size, overlaid with the total number of employees in each firm size category. While most firms employ 1 to 4 employees, the distribution of employees across firm sizes is more evenly spread.

discussion of the mechanisms behind the empirical findings.

The model's market structure consists of J formal firms that decide how many workers to hire from two types. Firms hire formal workers F paying payroll taxes and informal workers I "off the books" to avoid paying the payroll taxes, as in Ulyssea (2018).[20] Natives and migrants are perfect substitutes for informal labor $I = I^N + I^M$, and each firm $j = \{1, ..., J\}$ posts a pair of wages (w_{I_j}, w_{F_j}) that all workers i observe and decide to accept. Importantly, each firm has different work environments, measured by amenities a_{L_j}, and workers have idiosyncratic preferences ν_{i,L_j} depending on the labor group to which they belong $L \in \{I, F\}$. This yields a worker-specific job valuation at each firm, implying that firms face upward-sloping labor supply curves, which they internalize when choosing posted wages to maximize profits.[21] Online Appendix A presents the worker utility specification and the resulting firm-specific labor supply functions.

The profit function of firm j depends on a productivity shifter T_j, a price of the good P_j, and a production function Q_j:

$$\max_{I_j, F_j} \pi_j = P_j T_j Q_j - \tau(I_j) w_{I_j}(I_j) I_j - (1 + \tau_F) w_{F_j}(F_j) F_j. \tag{1}$$

Here, $\tau(I_j)$ represents a convex cost that is increasing in the firm's informal labor size. These convex costs are important for matching the stylized fact that informal labor decreases with firm size (Ulyssea, 2018) and for capturing the cost of evasion related to government law enforcement. Particularly, I assume that $\tau(I_j) = I_j^\eta$ with $\eta \geq 1$. The τ_F represents the constant payroll taxes firms must pay for formal workers, and the production function takes a constant elasticity of substitution (CES) form: $Q_j = (\alpha_I I_j^\rho + \alpha_F F_j^\rho)^{\frac{1}{\rho}}$, where $\alpha_I + \alpha_F = 1$ and α captures relative productivity weights. Thus, formal and informal workers can possess different attributes and, therefore, are imperfect substitutes, with an aggregate elasticity of substitution common across all firm types given by $\sigma = \frac{1}{1-\rho}$. To close the model, P_j is the inverse demand function defined as $P_j = D_j (T_j Q_j)^{-(1-\varepsilon)}$,

[20] In this model, I abstract from the decision of the firm to become formal or informal, as I focus only on the labor choices of formal firms. Moreover, I do not consider the general equilibrium effects of potential workers' transitions between the formal and informal sectors.

[21] For instance, preferences for working in a firm may refer to distance to the workplace or interactions with coworkers.

where $\varepsilon^D = -1/(1-\varepsilon)$ is the elasticity of product demand and D_j is the firm-specific product demand.[22]

In this partial equilibrium framework, I study the effect of an immigration shock that shifts the aggregate informal labor supply outward ($d\mathcal{J}$), consistent with the fact that around 90% of immigrants are informally employed. I examine the firms' response across the wage and employment margin, so the wage and employment elasticity for each type of worker in firm j is $\varepsilon_{w_{L_j},\mathcal{J}}$ and $\varepsilon_{L_j,\mathcal{J}}$, respectively. Differential firm-level responses to an immigration shock are a novelty of this framework. Unsurprisingly, in Online Appendix A I show that the elasticity of informal labor is always positive ($\varepsilon_{I_j,\mathcal{J}} > 0$) and the elasticity of informal wages is always negative ($\varepsilon_{w_{I_j},\mathcal{J}} < 0$), regardless of whether informal and formal workers are close substitutes or not.

More interestingly, I show how formal wages and employment of each firm change in response to the shock:

$$\varepsilon_{w_{F_j},\mathcal{J}} = \Omega_j s_{I_j}(\varepsilon - \rho). \tag{2}$$

Here, $s_{I_j} = \frac{\alpha_I I_j^\rho}{\alpha_I I_j^\rho + \alpha_F F_j^\rho}$ is the relative contribution of informal work to production before immigrants arrive, and $\Omega_j = \frac{1}{\xi_{I_j}\xi_{F_j} - (\varepsilon-\rho)^2 s_{I_j}\beta_I s_{F_j}\beta_F}$ is a positive parameter.[23] Firstly, if informal workers are sufficiently close substitutes for formal workers (such that $\rho > \varepsilon$), then the elasticity of formal wages with respect to aggregate informal labor is negative.

The corresponding expression regarding formal employment is equal to:

$$\varepsilon_{F_j,\mathcal{J}} = \Omega_j s_{I_j}(\varepsilon - \rho)\beta_F. \tag{3}$$

The implications in terms of the substitution parameter (i.e., $\rho > \varepsilon$) hold similarly to those for formal wages, though the response is now adjusted by β_F. This is consistent with the Clemens and Lewis (2022) framework, where a high elasticity of substitution across immigrants and natives relative to the output elasticity yields negative employment effects. Importantly, as the relative con-

[22] I do not distinguish whether the produced good is tradable or non-tradable (Burstein et al., 2020). Besides, I exclude any spillover labor demand shock arising from migrants' consumption of goods and services (Galaasen et al., 2025).

[23] Online Appendix A defines ξ_{I_j} and ξ_{F_j} and shows that Ω_j is always positive.

tribution of informal workers to production increases ($s_{I_j} \uparrow$), the adjustment on formal employment and wages is more negative ($\varepsilon_{F_j,\mathcal{J}} \downarrow$) as long as informal and formal workers are close substitutes.[24]

In this model, for certain low-productivity firms, formal wages can be downwardly rigid due to the existence of a minimum wage (e.g., $\overline{w}_{F_{Min}} \geq w_{F_j}$), which results in a muted formal wage margin (i.e., $\varepsilon_{\overline{w}_{F_{Min}},\mathcal{J}} = 0$).[25] I find the formal employment elasticity is higher in absolute terms than for the firms where the minimum wage does not bind.

Overall, I point to two main predictions. First, when substitutability between formal and informal workers is high, an informal labor supply shock reduces formal wages and employment. Second, the extent to which the minimum wage is binding and the firm's production depends on informal labor determines how responsive the firm is to such shocks. Specifically, firms that rely more heavily on informality for production will adjust formal wages and employment more intensely in response to an immigration shock. Additionally, if they pay their formal workers close to the minimum wage, the formal employment response becomes even more important.

In the model, a firm's informal production share is inversely related to its size. Large firms face higher marginal costs of hiring additional informal workers because of their convex costs. Therefore, this motivates the empirical analysis of worker responses across the firm-size distribution to validate the model's predictions. I also extend the worker-level analysis to worker responses across firms' productivity using the estimated AKM firm effects, as these are also linked to the firm's informal production share. Table 1 in Section 6 complements this exercise by directly measuring changes in the share of informality across firm-size categories using the labor force survey.[26]

[24]Due to the absence of educational information in the administrative data and to maintain broader predictions, the model abstracts from skill heterogeneity and focuses solely on the formal-informal distinction. While this approach prevents the analysis of differential impacts on high- and low-skilled natives within each sector, my data lack clear empirical counterparts to test such additional predictions.

[25]In Online Appendix A, I derive the remaining elasticities under this condition for those firms defined as demand constrained (j_m).

[26]The model focuses on adjustments in labor inputs within formal firms, where the coexistence of formal and informal labor allows changes in relative labor costs to affect input choices (Corbi et al., 2021; Kleemans and Magruder, 2018). Other channels may also contribute to the observed responses. First, when formal and informal firms compete in product markets, lower labor costs at informal firms may translate into lower prices and larger market shares, reducing demand for formal firms' output and labor (Rozo and Winkler, 2021; Bandiera et al., 2023). The available data do not allow me to test this price and market-share channel directly. Second, formal firm entry and exit can affect employment beyond the adjustments within firms captured by the model (Gulek, 2024). Third, worker reallocation and production networks may transmit effects across firms (Akgündüz et al., 2024; Gyetvay and Keita, 2023). The

4 Empirical Strategy

To quantify the evolving impact of immigration on worker outcomes and across firms, I estimate the following dynamic difference-in-differences (DiD) specification from $t = \{2012, .., 2019\}$, estimating separate yearly regressions of the following form:

$$\Delta Y_{i,l,t} = \alpha_t + \theta_t \Delta M_{l,2018} + X_i' \beta_t + \Delta u_{i,l,t}. \tag{4}$$

Here, the outcome $\Delta Y_{i,l,t}$ is the difference in wages, employment, or earnings within workers' pre- and post-treatment years relative to a baseline year. The immigration shock $\Delta M_{l,2018}$ varies for each local labor market l, and the vector X_i contains individual characteristics in 2015. Namely, dummies for six age groups interacted with dummies for sex and self-employment.[27] Broadly, this specification is akin to Foged and Peri (2016) regression with individual fixed effects, and compares individuals with similar observables in the base period but who were working in local labor markets with different exposure to the immigration shock, which I will describe below in detail. Hence, as I follow worker outcomes over time, θ_t measures the worker-level response to migration, with 2015 as the baseline year and $\theta_{2015} = 0$ by construction. By taking differences, I net out any individual constant unobservables. Lastly, the intercept for each year is α_t, and I cluster the standard errors across all specifications at the treatment level, which is the FUA (equal to $G = 109$).

The individual outcomes are more precisely defined as follows. First, the employment outcome is $e_{i,l,t} - \sum_{k=2013}^{2015} e_{i,l,k}/3$, where $e_{i,l,t}$ is the indicator of formal sector employment for worker i in local labor market l in period t, and the coefficient measures changes in the probability in pp. As in Yagan (2019), I use average employment in the pre-shock period to transparently account for workers' varying labor trajectories in the formal sector. In the event study figures, however, I use the simple difference with the base period ($e_{ilt} - e_{il,2015}$) to avoid pre-treatment coefficients being mechanically close to zero. Second, the wage outcome is $\frac{w_{i,l,t} - w_{i,l,2015}}{w_{i,l,2015}}$, and the coefficient

evidence below on worker mobility speaks to some reallocation margins, but does not rule out these other channels.

[27] Education and occupation data are unavailable in the PILA. Due to measurement errors, industry information is used as a robustness check rather than a baseline, since these codes were not verified until 2019.

measures the percentage change in wages for each worker with respect to 2015, so the worker must be observed both in 2015 and t. Third, the earnings outcome is $\sum_{t=2016}^{t=2018} \frac{Earnings_{it}}{Earnings_{i,2015}}$ and it measures changes in the evolution of earnings normalized by the earnings in the pre-shock period, following Autor et al. (2014). Yearly earnings are mechanically zero if the worker is not employed in the formal sector that year, whereas if employed, they equal the worker's wages. This outcome reflects the combined effect of the observed changes in employment and wages during the post-treatment periods. Because earnings are recorded as zero for workers who leave formal employment, these estimates should be interpreted as an upper bound on native workers' true income losses: workers displaced from the formal sector may retain some earnings in the informal sector that are not captured in the administrative data.

The immigration shock $\Delta M_{l,2018}$ is defined as:

$$\Delta M_{l,2018} = \frac{L_{Ven,l,post}}{L_{l,2018}}, \tag{5}$$

where the numerator is the stock of employed migrants from Venezuela (aged 18 to 64) in local labor market l who arrived in Colombia between 2016 and 2018. Employed migrants include both Venezuelans and returning Colombians. The denominator $L_{l,2018}$ is the total employed population in the local labor market. I mainly focus on and interpret the 2018 coefficient in the regressions (i.e., θ_{2018}) to match the census year and avoid rescaling the shock as for the coefficients of other periods. In this setting, both the numerator and denominator are potentially endogenous, which motivates the use of an instrumental variables strategy. As a robustness check, I construct the immigration shock using the 2005 employed population in the denominator and obtain quantitatively similar results (see Online Appendix Table F.6, row 6), indicating that the choice of denominator does not drive the findings. Lastly, this constant immigration shock is useful because it leverages the full census count rather than using other time-varying migration shares from surveys, and it allows for transparent placebo tests of pre-trends within the same analysis (Delgado-Prieto and Stuhler, 2026).

Migrants self-select into areas with better economic opportunities, so the immigration shock $\Delta M_{l,2018}$ is likely endogenous and its effects are likely upward-biased. To consistently estimate the effect of immigration on the outcome variables, I instrument the immigration shock $\Delta M_{l,2018}$ using the distance to the nearest bridge to Venezuela and past Venezuelan settlements. Both instruments are standard in this literature: distance-to-border and past-settlement instruments have been used throughout studies of the Venezuelan exodus (Bonilla-Mejía et al., 2024; Caruso et al., 2021) and Syrian refugees in Turkey (Del Carpio and Wagner, 2015; Ceritoglu et al., 2017; Aksu et al., 2022). The motivation for the IV approach is the following.

Distance Instrument. Distance plays a crucial role in the Venezuelan immigration, given that Colombia and Venezuela share a 2,220-kilometer land border. As a result, individuals' entry into the local labor market l is affected by the travel distance between the two countries, which imposes both time and economic constraints on immigrants. The distance-based component of the first stage includes linear and quadratic terms in the air-kilometer distance d_l to the nearest main crossing bridge with Venezuela.

A potential challenge to this instrument is that border areas may face additional economic shocks (such as reduced trade or business transactions) compared with areas farther from Venezuela, potentially violating the exclusion restriction. Online Appendix Figure F.4a offers suggestive evidence that the trade shock induced by the Venezuelan crisis began several years before the immigration shock. In particular, exports from border departments to Venezuela remained consistently close to zero post-treatment, so any potential threat must stem from long-lasting lagged effects. Still, I observe insignificant employment and wage effects in the firms with more than 100 workers, which are more likely to be exporters affected by trade shocks. I also show in Figure F.4b that trends in log GDP for border and non-border departments were similar before the immigration shock, indicating that any trade-related impact on broader economic activity was limited.[28] This

[28]To test this more formally, I estimate a dynamic specification at the department level, given data availability, using log exports to Venezuela and log GDP as outcomes. The pre-treatment coefficients on log exports are insignificant, though with expectedly larger values in 2012 and 2013 when trade with Venezuela was still active, while the coefficients on log GDP are close to zero throughout the pre-treatment period (see Online Appendix Figures F.6a and

is corroborated by the individual-level outcomes themselves: Figure 2a shows no differential pre-trends in employment or wages between more- and less-exposed areas before the migration shock, indicating that border areas were not already experiencing the effects of the broader Venezuelan collapse, through smuggling, trade, or exchange-rate arbitrage. Lastly, given that border areas also exhibit higher rates of immigrant regularization (Bahar et al., 2021), I exclude them to reduce potential confounding. The resulting point estimates remain similar, although wage effects become statistically insignificant.

Past-Settlement Instrument. The other instrument follows the classic methodology of Altonji and Card (1991) and Card (2001), using past Venezuelan settlement patterns to predict new arrivals. It is defined as:

$$z_l = \left(\frac{Ven_{l,2005}}{\sum_l Ven_{l,2005}} * L_{Ven,post} \right) / L_{l,2005}, \tag{6}$$

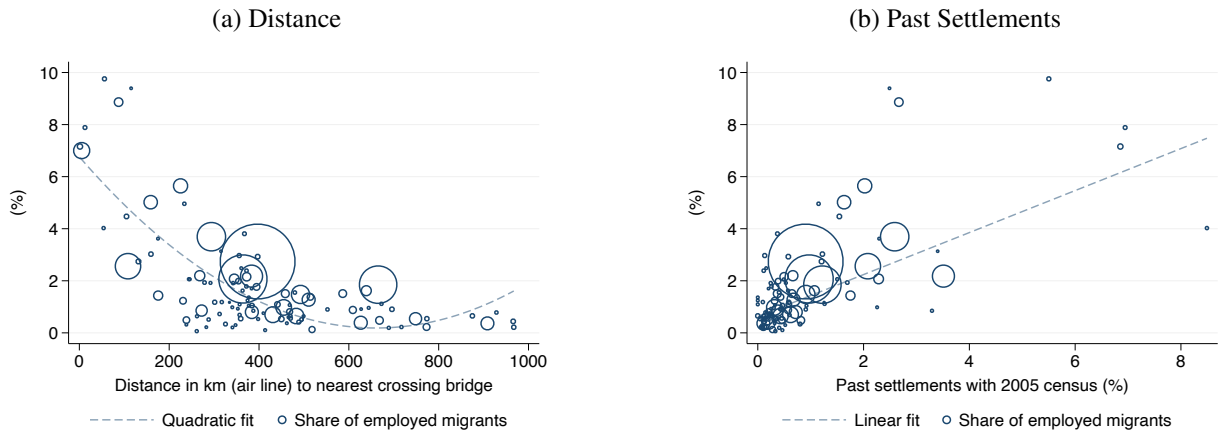
where the first term represents the share of Venezuelans in FUA l, according to the 2005 population census, normalized by the working-age population $L_{l,2005}$ in l at 2005, whereas $L_{Ven,post}$ is the number of employed migrants in Colombia who arrived between 2016 and 2018 according to the census. Since I have only one origin country, this corresponds to a just-identified shift-share instrument, where exogeneity relies on the share of Venezuelans (Goldsmith-Pinkham et al., 2020). The motivation for using past settlements as an instrument is that newly arriving immigrants tend to relocate to areas with established Venezuelan communities. To mitigate concerns about exclusion restrictions, I select a sufficiently lagged share in 2005 that captures persistent network effects without being systematically linked to current labor-demand shocks. A department-level regression testing pre-trends in log GDP and log exports to Venezuela with a similar instrument yields coefficients near zero for both outcomes (see Online Appendix Figures F.6a and F.6b).

First Stage. Figures 1a and 1b depict graphically the first stage of the immigration shock $\Delta M_{l,2018}$ for the 109 FUAs included in this analysis (see Online Appendix Map F.5 for the geographic distribution of the shock). These figures highlight the relevance and functional form of the instruments.

F.6b).

For the first instrument, a larger distance from a crossing bridge decreases the immigration shock in the FUAs until a point at which longer distances no longer imply lower immigration shocks, causing the curve's slope to bend upward. Based on past settlements, the second instrument shows a positive and almost linear relationship with the immigration shock. The immigration shock at the FUA level is substantial, with some areas experiencing an increase in the share of employed migrants from 7% to 10% of their overall employed population. An improvement over recent studies in Colombia is that I construct more granular local labor markets using administrative data, whereas the labor force survey is only representative of 24 departments.

Figure 1: Immigration Shocks and the Instruments



Note: I weigh dots by formal employment according to the PILA in 2015. In (b), I exclude one area to narrow the x-axis values. Functional Urban Areas in Colombia (G=109). Source: CNPV, 2018.

This paper aims to estimate the impact of immigration at the individual level, so the first stage of the two-stage least squares (2SLS) will weight each FUA by the number of individual observations available. With this in mind, the first-stage model at the FUA level is $\Delta M_{l,2018} = \delta + \gamma_1 d_l + \gamma_2 d_l^2 + \gamma_3 z_l + \upsilon_l$, where d_l is the distance to the nearest crossing bridge, z_l represents past settlements of Venezuelans, and γ_1, γ_2, and γ_3 are the first-stage coefficients. The error term υ_l captures the endogenous component of $\Delta M_{l,2018}$. In rows 1–3 of Online Appendix Table F.6, I estimate the main outcomes using both instruments jointly and separately. The results for the main outcomes using only the distance instrument yield slightly more negative coefficients than

those obtained with both instruments, though the estimates are noisier. Using only past settlement instruments, the coefficients are also more negative than when combining both. The decision of which instruments to choose embeds a trade-off between different exclusion restrictions and the relevance of the combined instruments.[29] Given that the estimates from each instrument are not substantially different, jointly they improve the precision of the estimates, and the R^2 in Online Appendix Table F.5 of the first stage regression is higher with both, although the F-statistic is lower, I combine them for the main analysis.

5 Worker-Level Responses

This section examines the impact of immigration on formal wages and employment at the worker level and then explores the heterogeneity of these effects across worker characteristics. One advantage of the dynamic DiD specification is the ability to test for differential trends in outcomes before the immigration shock occurs. Figure 2a shows no significant pre-trends for employment or wages, supporting the idea that workers in less immigration-exposed areas are a candidate counterfactual for the outcomes of workers in more exposed areas.

In the post-treatment periods, OLS coefficients are close to zero, presumably upward-biased, while the 2SLS regression helps mitigate this bias, resulting in more negative coefficients. Since pre-trends are relatively stable before the arrival of immigrants with both methods, this suggests that immigrants are arriving in areas more likely to have positive demand shocks in the post-treatment periods, probably due to improving economic opportunities, which biases the OLS estimates towards zero. For this reason, I only analyze the 2SLS coefficients from here on.

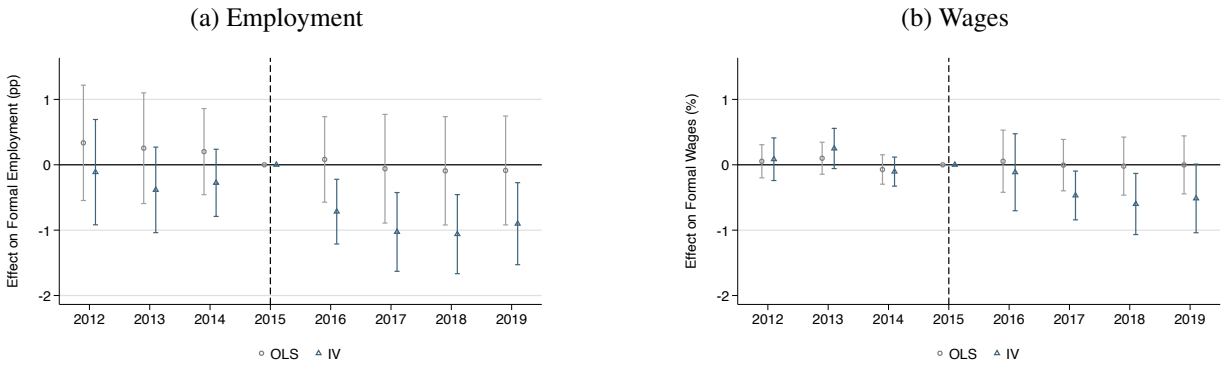
In 2018, the census year, a one pp increase in the immigration shock in an area reduces the probability of formal sector employment by 1.1 pp (see Figure 2a).[30] To express the magnitude in

[29] An over-identification test requires assuming one instrument's exogeneity to evaluate the other, and rejecting the null hypothesis might instead mean that each instrument identifies a different treatment effect.

[30] This regression uses $e_{i,l,2018} - e_{i,l,2015}$ as the dependent variable, capturing the change in the employment status from 2015 to 2018. For the heterogeneity analysis, the dependent variable is $e_{i,l,t} - \sum_{k=2013}^{2015} e_{i,l,k}/3$, which compares the post-shock employment status to the average pre-shock employment, yielding slightly less negative coefficients.

relative terms, I use as a benchmark the formal employment probability of 0.42 in the labor force survey for workers aged 25 to 55 in 2015. A 1.1 pp decline is equivalent to 2.6% of this survey benchmark. The difference in exposure between the 75th and 25th percentiles implies a 1.65 pp decline in formal employment probability, equivalent to 3.9% of the same benchmark.[31] In Figure 2b, I find a coefficient of –0.6% on formal wages in 2018 for a one pp increase in the immigration shock.[32] Consequently, a worker at the 75th percentile of exposure relative to one at the 25th percentile of exposure experiences a drop of 0.9% in formal wages. Thus, the impact on wages is minor compared to the impact on employment.

Figure 2: Event Study Estimates on Individual Wages and Employment



Note: I estimate Equation (4) separately by year. The sample is restricted to natives aged 25 to 55. In panel (a), there are 6,706,035 workers, while in panel (b), this varies slightly by year, as the worker must be employed in the post-treatment and base year. Controls include interactions of sex with six age categories and a dummy for self-employment in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA 2012–2019.

I then examine the heterogeneity in wage and employment estimates based on observed workers' characteristics prior to the immigration shock to show how immigration effects vary.[33] The coefficients for each subgroup come from separate regressions of the main empirical specification

[31] The 25th and 75th percentile migration rates are 0.6% and 2.1%, respectively. Thus, $(2.1-0.6)\times(1.1/0.42)\approx 3.9\%$ of the survey benchmark.

[32] In comparison to Osuna Gomez and Medina-Cortina (2026), which studies the deportation of migrants from the United States to Mexico, they find that a one pp increase in the deportation shock decreases the probability of formal employment among low-wage workers by about 0.11 pp and reduces formal wages by around 0.2%. These results are consistent with my findings, though the magnitudes are smaller.

[33] Online Appendix Table F.11 shows broadly flat pre-trends across most worker subgroups, although not all. These checks provide support for the identification assumption but do not rule out differential trends.

(4). First, Online Appendix Figure F.7 shows that self-employed natives, who can more easily exit the formal sector than employees with rigid labor contracts, experience a greater decline in formal employment following the immigration shock.[34]

Second, Online Appendix Figure F.8 shows that older workers experience a larger decline in the probability of formal sector employment than younger workers, with a less pronounced pattern for wages.[35] Online Appendix Figure F.10 extends this to labor market entrants and workers nearing retirement, with the highest negative employment effect among the oldest workers. Third, the impact on employment and wages is similar for men and women. Fourth, Online Appendix Figure F.11 shows that the employment shock is more severe for workers with lower job tenure, consistent with severance payments and firm-specific human capital making longer-tenured employees less substitutable by migrants.[36]

5.1 Impacts of Immigration Across the Wage Distribution

Because immigration can disproportionately affect natives with a specific set of skills, I estimate worker-level impacts across the baseline wage distribution, which serves as a proxy for education. For this analysis, I categorize native workers into seven bins based on their local wage distribution. Figure 3 illustrates the uneven effects of immigration: native workers earning the minimum wage and having the lowest skills within the formal sector experience the most negative impact on formal employment, while workers above in the wage distribution are unaffected. Specifically, a 1 pp increase in the immigration shock leads to a 1.5 pp decrease in the probability of formal sector employment for minimum-wage workers. Because the immigration shock considers all employed immigrants locally, whether formal or informal, I scale the coefficient by the size of the formal sector (around 0.5) and by the share of minimum-wage workers within the formal sector (approximately 0.4) to accurately benchmark the effects. This calculation indicates that one additional

[34] Labor income data for self-employed workers in PILA are noisy as it includes public and private contractors that typically report only 40% of their labor income by law. Nevertheless, the wage point estimates are more negative for the self-employed than for employees.

[35] Online Appendix Figure F.9a reports the corresponding earnings estimates by age group.

[36] Online Appendix Table F.7 further examines heterogeneity by combining baseline age and tenure.

employed immigrant displaces roughly (0.5*0.4*1.5=) 0.3 native minimum-wage workers in the formal sector. Due to the relatively high and binding nature of the minimum wage for around 40% of formal workers in the pre-shock period, these workers face a greater risk of job displacement.[37] Additionally, the presence of a large informal sector is relevant, as minimum-wage workers tend to be less educated and can more frequently transition into informality.[38]

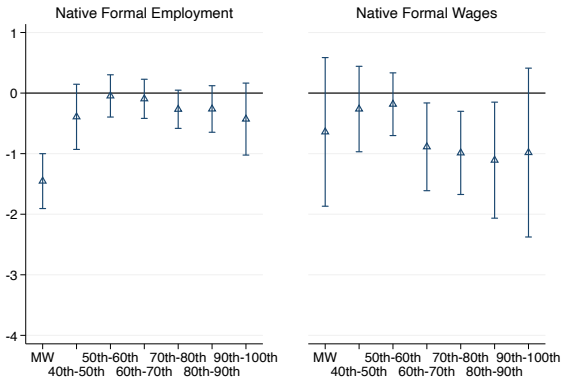
Conversely, formal workers who earn the minimum wage are the least affected by the immigration shock in terms of wages, partly due to their downward wage rigidity. In contrast, for workers between the 60th and 90th percentiles of the local wage distribution, who earn around two to three times the minimum wage on average, I observe a negative wage effect of 1%-1.2%.[39] Increased competition from mid- and high-skilled migrants entering the formal sector is a potential explanation for this wage decline. Online Appendix Table F.3 shows a sharp increase in other foreigners between 2015 and 2019, who often earn high wages. In addition, the contraction of the formal sector may strengthen the bargaining power of formal firms, which could affect the formal wages of incumbent workers earning above the minimum wage. This does not necessarily imply an absolute wage decrease. The coefficient reflects the average wage growth of native workers in areas with higher migration exposure compared to those with lower exposure. Lastly, these findings suggest that potential selection issues into the sample of workers employed in both 2015 and the comparison year used for the worker-level wage regressions may be minor. Employment responses are near zero above the minimum wage, leaving limited scope for differential attrition, while near the minimum wage, large employment effects make selection a more plausible concern, but wage rigidity limits any differential wage changes.

[37] Conditional on being employed in the two periods, around 75% of minimum wage earners still earn the minimum wage after three years.

[38] Online Appendix Section C shows corresponding survey-based evidence that regional employment responses in the informal sector are minor compared to those in the formal sector.

[39] Online Appendix Figure F.9b reports the corresponding earnings estimates by baseline wage category.

Figure 3: Employment and Wage Estimates by Individual Wage at Baseline



Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to natives aged 25 to 55. The dependent variables are employment relative to the pre-shock period and wages relative to the base period. Controls include interactions of sex with six age categories and a dummy for self-employment in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019 .

Threats to the Identification Assumptions. Even if border areas have similar trends before immigrants arrive as non-border areas, they may be more exposed to post-treatment shocks derived from the Venezuelan crisis, potentially violating the exclusion restriction. I address this concern in four different ways. First, by including a Bartik-type local labor demand control, measured as predicted local employment growth between 2015 and 2018 using baseline industry employment and national employment growth by industry, to capture latent demand shocks across areas. Online Appendix Figure F.12 shows that the results across the wage distribution remain essentially unchanged, suggesting that differential demand shocks are less of a concern in this setup. Second, using only past settlements as the instrument, Online Appendix Table F.6 shows the coefficients remain significant and are even more negative. Third, by excluding border areas from the analysis to further mitigate concerns that they confound the effects, Online Appendix Table F.6 shows more negative point estimates for employment and wages.[40] Fourth, after including controls for the baseline wage of the worker as a proxy for workers' education, I find similar results for wages and

[40]Online Appendix Table F.6 also reports robustness checks excluding Bogotá, which accounts for 35.3% of formal employment in the estimation sample, using real wages, and top-coding wages at the 99th percentile.

employment, with slightly more negative effects for wages. By further controlling for the baseline wage distribution, any remaining threat must arise from a specific shock that differentially affects workers with similar wages, age, gender, and job type in the more exposed areas.

Comparison of Worker-Level and Regional-Level Effects. Most of the migration literature focuses on regional responses when examining the effects of immigration shocks.[41] Since regional responses aggregate multiple margins of adjustment to immigration, they can yield findings that differ from worker-level responses, as documented by Dustmann et al. (2026). To address this, I adapt the employment decomposition they introduce to shed light on these different responses.[42] Specifically, I decompose the changes in regional formal employment into three different components: (1) a displacement of incumbent workers –outflows from formal employment–, (2) hiring of new formal workers or inflows from other regions –inflows to formal employment–, and (3) relocation of existing employed formal workers to other regions.

In this analysis, worker-level employment estimates capture the outflows or displacement of incumbent native workers from the formal sector. In contrast, the regional-level estimate from cross-sectional data combines all three adjustment margins. These two responses are complementary and address different policy questions. Online Appendix Figure F.13 shows the decomposition of the regional formal employment response at the FUA-level (–1.0%), breaking it down into the three components: outflows to non-employment or the informal sector (1.2%), inflows from other regions, non-employment or the informal sector (–0.7%), and relocation to other regions (–0.8%).[43]

[41] Recent regional-level studies include Monras (2020) in the US and Muñoz (2024) in the EU. The first study documents that low-skilled Mexicans who left their country due to the peso crisis had a high transitory impact on local labor markets in the US. The second study exploits a trade liberalization in services across Europe to find negative regional effects on the employment of domestic workers.

[42] The main distinction, relative to Dustmann et al. (2026), is that the outflows and inflows margins in this study can be decomposed further into non-employment and the informal sector. Unfortunately, there is no panel data for the informal sector to measure these decompositions.

[43] I estimate each component coefficient by running separate regional regressions of each share on the instrumented local immigration shock and use the local formal employment in the base period as regression weights. Displayed coefficients are rounded.

More formally, the employment decomposition equals:

$$\frac{E_{r1}-E_{r0}}{E_{r0}} = -\underbrace{\frac{E_{r,Out}}{E_{r0}}}_{\text{Outflows}} + \underbrace{\frac{E_{r,In}}{E_{r0}}}_{\text{Inflows}} - \underbrace{\frac{E_{r,Move}}{E_{r0}}}_{\text{Relocation}}. \tag{7}$$

In this analysis, the only significant margin is outflows from the formal sector, a finding that contrasts with Dustmann et al. (2026), where inflows are the most prominent margin. Some of the differences in findings can be attributed to the context: Colombia has a large informal sector, which enables firms to hire informally after displacing formal workers, and less restrictive job-protection regulations compared to Germany.

Regarding wage estimates, the worker-level response shows a 0.6% decrease, while the regional-level estimate in Online Appendix Table C.1 is a statistically insignificant 0.5% decrease. The point estimates are similar, and their different significance levels do not establish that the effects differ. As noted in Dustmann et al. (2026), worker-level wage regressions capture changes in the price of labor while holding constant the population composition. In contrast, regional-level wage regressions jointly capture changes in the selection and composition of workers due to inflows and outflows, as well as changes in labor prices. Compositional changes could nevertheless influence the regional estimate as follows. By mainly displacing minimum-wage workers, the immigration shock could positively select those who remain employed, mechanically raising average regional formal wages (see Figure 3). However, immigration lowers labor costs in certain mid- and high-wage subgroups, thereby lowering regional formal wages on average. These potentially counteracting effects illustrate why regional and worker-level estimates can capture different margins of adjustment. The estimates here do not separately identify these channels, but motivate examining both aggregate local labor markets and individuals within them.

Individual panel data also offer insights into inter-regional movements in response to immigration. For example, Foged and Peri (2016) document that younger workers in Denmark are more mobile following refugee arrivals. Online Appendix Table F.8 shows the effect on regional movements by age groups. Younger formal workers tend to move more, though the coefficients

are insignificant. While point estimates decrease with age, all remain insignificant, suggesting that mobility may be a less relevant adjustment margin in this context.

6 Worker-Level Responses Across Firms

The model suggests that formal workers in smaller and less productive firms are more responsive to the immigration shock. I turn to the data to empirically test these implications. The firm dimension is also key for this heterogeneity analysis for three reasons (Delgado-Prieto, 2024). First, over 80% of Venezuelan immigrants work in firms with fewer than 11 workers. Second, more than 50% of workers in the smallest formal firms earn the minimum wage and are thus on the margin of informality. Third, smaller firms employ a higher share of informal workers. Therefore, the exposure to the immigration shock varies substantially by firm size.

By comparing worker outcomes in small or large firms across local labor markets over time, I rule out time-varying effects of firm size from the impact of the immigration shock.[44] Figure 4 shows that native workers in firms with fewer than 50 employees in the pre-shock period experience the most negative effect on the post-shock probability of remaining in formal employment, whereas workers in larger firms are less affected. This is useful for showing transparently that trade shocks from the Venezuelan crisis are less of a concern in this context, as the main effects are observed in small, non-exporting formal firms, which are less affected by trade shocks. I also control for firms' industry and find similar, even more precise patterns, indicating that this is not driven by specific industries populated by small firms (see Online Appendix Figure F.14).

Workers in the smallest firms (those with fewer than 10 employees) face slightly more negative wage point estimates than workers in larger firms, but the patterns are less clear than those for employment.[45] This aligns with model predictions, where firms adjust wages less heavily in

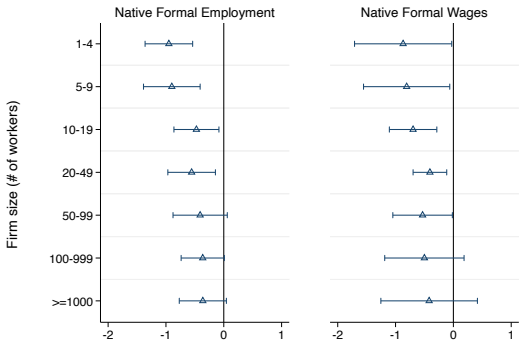
[44] I categorize workers by firm size in 2015, the year before the immigration shock, and examine worker-level employment and wage coefficients for 2018, the census year. Online Appendix Table F.13 shows broadly flat pre-trends across most firm-size subgroups, although not all, particularly for wages.

[45] Firm age is an alternative characteristic that can determine immigration impacts on workers. Online Appendix Figure F.15 presents results based on the worker's pre-shock firm age, and Online Appendix Table F.9 further separates this heterogeneity by firm age and size jointly, motivated by Fort et al. (2013).

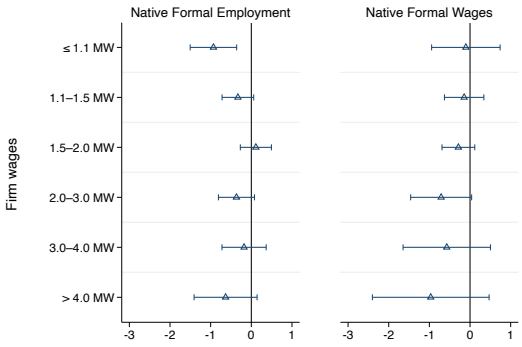
response to an immigration shock due to the presence of minimum wages and labor market power. In particular, Figure 4 is consistent with employment absorbing more of the adjustment when minimum-wage constraints limit wage reductions (Bhuller et al., 2025).

Figure 4: Employment and Wage Estimates

(a) Firm Size



(b) Firm Wages Relative to the MW



Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to native employees between 25 and 55 years old. Dependent variables are employment relative to the pre-shock period and wages relative to the base period. I use as controls the interactions of sex with six age categories. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Workers are observed in August of each year. Coefficients report employment (in percentage points) and for wages (in percent) for a one-percentage-point increase in the immigration share. Source: PILA, 2013–2019.

Table 1 provides direct survey-based evidence for the mechanism of increased informality among small firms: the negative effect on formal employment is largest for firms with 1–9 workers (-6.2%) and 10–19 workers (-5.4%), declines with firm size, and becomes insignificant for firms with more than 100 workers, while the informality share rises significantly in firms with 10–100 workers. This firm-size gradient closely mirrors the administrative-data pattern above and is consistent with the model's prediction that smaller firms have more elastic demand for formal labor.[46]

[46]Some caveats apply to these estimates, which are based on cross-sectional survey data for 24 departments, where firm size is self-reported by workers. As such, changes in employment by firm-size category may reflect both within-firm workforce adjustments and compositional shifts in the pool of employed workers across firm-size bins.

Table 1: Formal Employment and Informality Share by Firm Size

	Aggregate	1-9	10-19	20-100	100+
	(1)	(2)	(3)	(4)	(5)
Panel A: Formal employment					
β_{pre}	-0.237	0.333	1.383	2.822	-0.920
	(0.736)	(1.448)	(1.402)	(1.854)	(0.876)
β_{2018}	-1.921^{***}	-6.213^{***}	-5.398^{***}	-4.025^{**}	-0.364
	(0.473)	(1.498)	(0.721)	(1.842)	(0.634)
Panel B: Informality share					
β_{pre}	0.050	0.214	0.789	0.079	0.187^{**}
	(0.074)	(0.314)	(1.012)	(0.695)	(0.089)
β_{2018}	0.551^{***}	0.342	0.644^{*}	0.790^{*}	0.100
	(0.191)	(0.262)	(0.332)	(0.403)	(0.136)

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Regressions are estimated at the department level for 24 departments from 2011 to 2019. Panel A reports changes in log formal employment for all workers aged 18 to 64 in urban areas, both in the aggregate and by four firm-size categories (1–9, 10–19, 20–100, and 100+ workers). Panel B reports changes in the share of informality. For the firm size categories, I further restrict to salaried workers. β_{pre} reports the average of the pre-treatment coefficients (2011–2014). The immigration shock $\Delta M_{d,2018}$ is instrumented with the distance to the nearest crossing bridge with Venezuela and past Venezuelan settlements as in Delgado-Prieto (2024). Firm size is defined by the number of salaried workers. Standard errors are clustered at the department level. Coefficients report employment (in percent) for a one-percentage-point increase in the immigration share. Source: GEIH, 2011–2019.

Job-to-job Flows. A complementary way to test for changes in labor choices relies on administrative data. For this exercise, I construct a proxy that captures how closely formal firms are connected to the informal sector. The key idea is to use firms' hiring patterns to infer their desirability ranking in the formal labor market from job-to-job flows. Particularly, I develop an insider index inspired by the poaching index in Bagger and Lentz (2019), which classifies firms by the source of their new hires. The intuition is as follows: some formal firms mainly hire workers with formal-sector experience, while others predominantly recruit workers new to the formal sector. The latter group consists largely of labor market entrants or workers transitioning from informality. Firms in this second group effectively serve as entry points into the formal sector.

Formally, the index measures the share of a firm's hires in year t that come from within the formal sector:

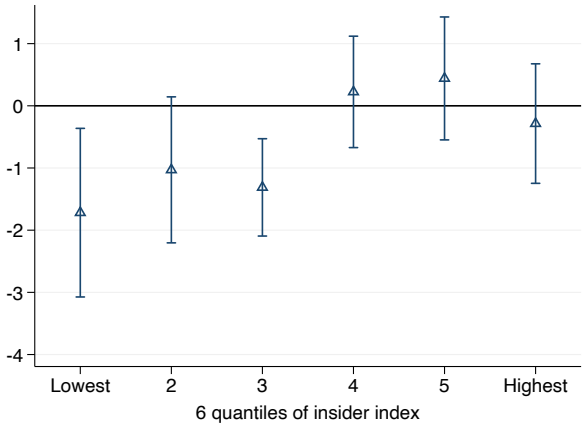
$$\kappa_{j,t} = \frac{N_{j,t}^{In}}{N_{j,t}^{In} + N_{j,t}^{Out}}, \tag{8}$$

where $N_{j,t}^{In}$ denotes the number of workers hired by firm j in year t who were previously employed in the formal sector, and $N_{j,t}^{Out}$ denotes hires from outside the formal sector.[47] A higher value of $\kappa_{j,t}$ indicates that a firm primarily recruits experienced formal workers. I then compute changes in the insider index between 2015 and 2018, $(\kappa_{j,2018} - \kappa_{j,2015})$, and assign these changes to workers based on their employer in each year. This allows me to study how firms' hiring composition responds to the immigration shock.

Figure 5 presents results across six quantiles of the pre-shock insider index. Firms that were initially more reliant on hiring from outside the formal sector experience a sizable decline in their insider index after immigration. A 1 pp increase in the migration rate reduces the insider index of firms in the lowest quantile by about 1.7 pp. In contrast, firms that primarily hire from within the formal sector exhibit little change. These patterns suggest that the formal firms more closely linked to the informal sector hire relatively fewer formal-sector workers following the immigration shock, potentially indicating they are hiring more from the informal sector.

[47] Hiring data are available from 2007 to 2018, observed in February and August. The index is undefined for firm–years with no hiring.

Figure 5: Index Estimates by Quantiles of the Baseline Insider Index



Note: Reported coefficients are θ_{2018}, measuring changes in 2018 relative to 2015. Dependent variable is the change in the insider index for workers employed in firm j in the base period between 2015-2018. Controls include interactions of sex with six age categories in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Workers are observed in August of each year. Source: PILA, 2013–2019.

Firm Exit and Entry. A supplementary explanation for previous employment findings is the lack of formal firm dynamism, particularly regarding the opening of new formal firms. Hence, I estimate the likelihood that firms hire formal workers entirely after the immigration shock. For this exercise, I count the number of firms in each area over time and perform regional-level regressions, excluding self-employed workers. Table 2 shows that formal firms experience negative growth in areas receiving more immigrants compared to those receiving fewer immigrants, though the coefficient is insignificant at the 5% level. When this growth is broken down into exits and entries, there is a marked increase in firm exits. A 1 pp increase in immigration increases the firm exit rate by 1.2%. This does not necessarily mean that firms cease operations entirely, as they may continue to hire all workers informally. In contrast, the firm entry rate remains almost unchanged, suggesting that immigrants may not be spurring the creation of formal firms in the short term (up to 3 years). This pattern can be consistent with intensified price competition in the product market from informal firms facing lower labor costs (Rozo and Winkler, 2021; Bandiera et al., 2023).

Table 2: Decomposition of Firm Growth

	(1) Total Firms	(2) Firm Exit	(3) Firm Entry
$\Delta M_{l,2018}$	-1.127	1.190*	0.063
	(0.750)	(0.582)	(0.935)
N	109	109	109

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Reported coefficients are θ_{2018}, measuring changes in 2018 relative to 2015. Regressions are estimated at the regional level for 109 FUAs weighted by their formal employment in 2015. The outcome variable in (1) is the percent growth in the number of firms, while in (2) and (3), I decompose the percent growth in terms of the exit and entry of firms, respectively. The sample is restricted to firms with at least one native employee, excluding self-employed workers. Firms are observed in August of each year. Source: PILA, 2015–2018.

Heterogeneity by Firm Pay Premiums. The model indicates that firm productivity is an important predictor of the effects of immigration. Thus, with access to the job matches of formal workers and firms in Colombia, I can construct a measure of firms' wage premiums, serving as a proxy of their productivity (Card et al., 2018). To achieve this, I estimate the standard AKM model proposed by Abowd et al. (1999), which decomposes the contribution of firm-specific and worker-specific constant characteristics to log formal wages $\ln w_{it}$. The AKM model is:

$$\ln w_{it} = \alpha_i + \psi_{j(i,t)} + X_{it}'\beta + \varepsilon_{it}. \tag{9}$$

Here, α_i captures the unobserved worker effect, ψ_j captures the unobserved firm effect, and $j(i,t)$ refers to the firm j where worker i is working in t. X_{it} is a vector of controls: age squared and cubic after normalization, plus year FEs, and ε_{it} is the error term. To rule out possible endogenous workers' movement due to the immigration shock, I estimate the model using data from 2010 to 2015 for August, restricting the sample to the largest set of firms connected by workers' mobility.[48]

Online Appendix Table B.2 presents the decomposition of the variance of wages $Var(\ln w_{it})$ in Colombia's formal sector using the leave-out method of Kline et al. (2020). Worker effects explain 50.1% of the variance of wages, and firm effects explain 15.7%, in line with the literature cited

[48] ψ_j is identified through workers' movements across firms, assumed exogenous conditional on worker and firm fixed effects (i.e., $E[\varepsilon_{it}|\alpha_i, \psi_j(i,t), X_{it}] = 0$).

in Card et al. (2018). Compared with middle-income countries, the share of variance explained by firm effects is lower in South Africa and Mexico and similar to that in Brazil (Kline, 2024). A potential explanation is Colombia's relatively high and binding minimum wage, which compresses the firm wage policies of lower-paying firms, decreasing the importance of firm effects. Lastly, the positive sorting of high-wage workers into high-wage firms accounts for another 21.6% of the variance. In four European countries and the US, this sorting explains between 10% and 20% of the wage variance (Bonhomme et al., 2023).[49]

Using the estimated $\hat{\psi}_j$, which are relative to the largest employer in the country, I now categorize workers into seven quantiles of firm FEs, from lowest- to highest-paying firms, to analyze heterogeneity.[50] For this exercise, the typical concerns about limited mobility bias affect only the precision, not the consistency of $\hat{\psi}_j$ (Bonhomme et al., 2023).[51] With this in mind, Figure 6 shows that workers at lower-paying firms experience negative employment effects while having insignificant wage changes. This contrasts with workers at higher-paying firms who do not experience negative employment effects.[52] A possible explanation is that the share of firms in the low-pay sector grows as immigrants predominantly work in these firms. Consequently, high-pay sector firms may extract higher rents from workers, thus reducing their wages, as shown in a model with on-the-job search in Amior and Stuhler (2022).

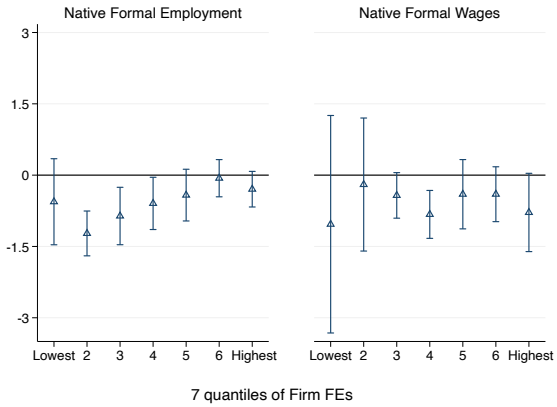
[49]The four European countries are Austria, Italy, Norway, and Sweden. The method they use for estimating the sorting in 6-year panels is the correlated random effects based on the grouping proposed by Bonhomme et al. (2019).

[50]Online Appendix Figure F.16 presents a similar analysis but dividing workers into seven quantiles based on worker FEs $\hat{\alpha}_i$.

[51]Online Appendix Table B.1 reports employment and wage descriptive statistics by quantile, noting that the smallest firms can be absent from this analysis because of the largest connected set restriction.

[52]To determine whether employment or wage changes prevail, I estimate the earnings outcome across quantiles of firm FEs. Online Appendix Figure F.9c shows that workers in the lowest-paying firms experience a more pronounced decline in earnings than workers from middle- to high-paying firms.

Figure 6: Employment and Wage Estimates by Quantiles of Firm FEs



Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to native employees between 25 and 55 years old. The dependent variables are employment relative to the pre-shock period and wages relative to the base period. I compute Firm FEs in the first stage using the standard AKM framework, with age squared and its cubic as time-varying controls, for the period 2010-2015. I use as controls in the second stage interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019.

Worker Sorting. Recent papers indicate that workers' reallocation across firms is a relevant response to labor market shocks. For example, Germany's introduction of a national minimum wage led low-wage workers to reallocate to higher-paying firms (Dustmann et al., 2022). I examine the reallocation effects of an immigration shock by analyzing changes in the sorting of high- and low-paying workers into high- and low-paying firms, which could provide a complementary explanation for the observed negative wage effects.[53] I construct the outcome using $\hat{\psi}_j$ values from Equation (9) and exploiting the movements of workers between firms in the post-treatment period. More concretely, the outcome measures the change in the AKM firm FEs in 2018 relative to 2015: $\hat{\psi}_{i,\{j=2018\}} - \hat{\psi}_{i,\{j=2015\}}$. The difference is zero if the worker remains in the same firm during that period.[54]

[53]For France, Orefice and Peri (2024) finds that high-paying workers are moving more into high-paying firms following the arrival of migrants. Similarly, Gyetvay and Keita (2023) finds that German natives shift from low- to high-paying firms.

[54]Firms' FEs are based on the pre-policy period, excluding firms created post-2015. The estimated firm FEs are adjusted to positive values for the outcome.

To determine whether low- or high-wage workers are sorting into higher-paying firms after the immigration shock, I present results across seven quantiles of worker FEs. A positive coefficient suggests a positive sorting effect from immigration. Online Appendix Figure F.17a shows that none of the categories yield significant results, indicating no differential sorting due to immigration.[55] Therefore, the negative wage coefficient observed in workers from high-paying firms likely reflects lower wage growth within these firms due to increased competition from migrants. Furthermore, Online Appendix Figure F.17b shows there is no evident reallocation of workers between larger or smaller firms post-immigration shock.

7 Heterogeneous Treatment Effects With Machine Learning

In this section, I move beyond ad hoc subsample splits and adopt a data-driven approach to systematically uncover heterogeneity in treatment effects.[56] I use machine learning to identify the subgroups most affected by immigration and the variables predicting these differences. Specifically, I use the approach proposed in Athey and Imbens (2016) and generalized in Athey et al. (2019), which was recently implemented by Gulyas and Pytka (2026) and Yakymovych et al. (2022). This framework identifies the subgroups experiencing the greatest wage and employment losses through a recursive partitioning method that allows for non-linear effects and high-order interactions between firm and worker variables. I use the generalized random forest (GRF) method from Athey et al. (2019) to build causal forests in the spirit of random forests (Breiman, 2001) but splitting the data according to a criterion on treatment effect heterogeneity.[57]

The procedure in Athey and Imbens (2016) and Athey et al. (2019) for building causal trees involves multiple steps, which I adapt to this setup. I explain the algorithm, the main specification, and how to construct the resulting conditional average treatment effects (CATEs) in Online Ap-

[55] This lack of significant sorting may be due to the macroeconomic conditions in Colombia, where unemployment slightly increased during the study period.

[56] In Appendix D, I present a conventional heterogeneity analysis that shows the negative effects of immigration on employment and wages vary by the intersection of worker and firm characteristics.

[57] I use the `grf` package in `R` based on Athey et al. (2019) to estimate the causal forests.

pendix E. To mitigate overfitting concerns, I estimate the causal forest using 2,000 decision trees with a minimum node size of 300 while clustering observations by FUAs.[58]

Identifying Most Affected Workers. I use the CATEs estimated by the algorithm to identify which subgroups are most affected by immigration. For this exercise, I divide native workers into quintiles based on their treatment effects for employment and wages (Q1 represents the most negative effect, while Q5 represents the most positive effect). The goal is to compare characteristics between these quintiles rather than making inference from the predicted CATEs. Accordingly, I do not report standard errors or hypothesis tests on individual CATEs or on differences across quintiles, since the quintiles are themselves constructed from the estimated treatment effects, and formal confidence intervals on them would also need to account for first-stage estimation uncertainty. Using subgroup rankings to characterize heterogeneity follows standard practice in the growing literature applying causal forests to labor market outcomes (Gulyas and Pytka, 2026; Yakymovych et al., 2022).

Online Appendix Tables F.10a and F.10b provide an overview of worker and firm characteristics in the pre-shock period. Native workers experiencing the most negative employment effects are generally older, have the lowest job tenure, and earn the lowest baseline wages. These workers are also employed in the smallest firms and the lowest-paying firms. In contrast, those facing the most negative wage effects are relatively younger and earn the highest baseline wages. The smallest firms also employ these workers, but in terms of pay premiums, they work in middle- to high-paying firms.[59] From a policy perspective, having the distribution of CATEs is useful for designing targeted measures to mitigate the adverse impacts of immigration in the most affected subgroups.

To better illustrate which subgroups are most affected by immigration, I construct heat plots that visualize the predicted conditional average treatment effects stratified by baseline wages and

[58] I set the tunable parameters of the algorithm to their default values, including honest splitting, and chose a relatively small minimum node size of 300 for precision. In further cross-validation exercises, results hold when I substantially increase the minimum node size.

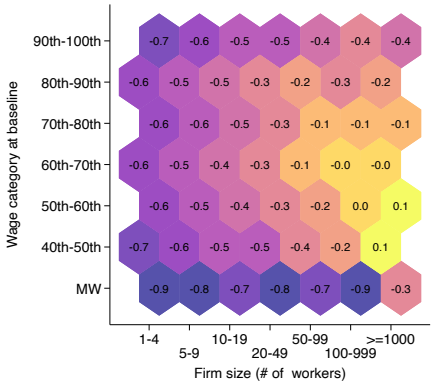
[59] In Online Appendix Figures F.18a and F.18b, I verify that the quintiles of treatment effects from the causal forest align with the main regression. Notably, the estimates follow the same order across both wages and employment.

firm size. Figures 7a and 7b show an overview of the effects across these two dimensions. The greatest negative impact on employment is concentrated among minimum-wage earners in small and medium-sized firms. In contrast, the most negative wage impacts are concentrated among high-wage workers and gradually disappear as the worker's firm grows. Hence, both employment and wages have the greatest negative impacts in smaller firms, but differ across workers' wage distributions, complementing previous results.

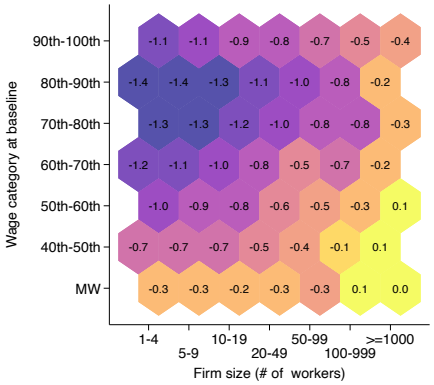
Figures 7c and 7d then display average treatment effects by quantiles of firm FEs intersected with quantiles of worker FEs. The most negative employment effects are concentrated among low-wage workers in the lowest-paying firms. Conversely, the most negative wage effects tend to be concentrated among high-wage workers in low- to middle-paying firms.

Figure 7: Heat Plots of Conditional Average Treatment Effects by Worker and Firm Characteristics

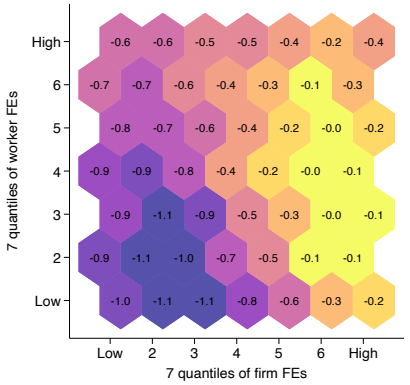
(a) Employment: Worker Wage Category × Firm Size



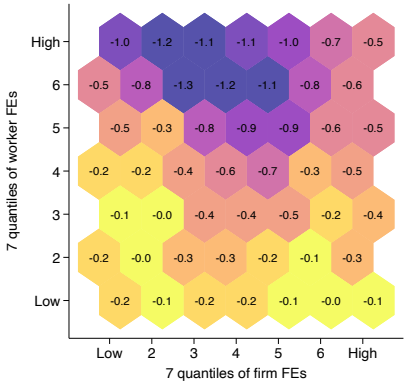
(b) Wages: Worker Wage Category × Firm Size



(c) Employment: Worker FEs × Firm FEs



(d) Wages: Worker FEs × Firm FEs



Note: The forests use 2018 wage changes relative to 2015 and employment changes relative to average employment in 2013–2015. Each hexagon reports the mean predicted conditional average treatment effect within a subgroup defined by worker and firm characteristics, estimated using a causal forest. Outcomes are changes in employment or wages in 2018 relative to the pre-shock period, with predicted immigration share as the treatment. The sample is restricted to natives aged 25 to 55. Standard errors are clustered at the FUA level. The causal forest is trained on 50% of the main sample.

Variable Importance for Heterogeneity. An approach to summarize these findings involves using a variable importance measure commonly employed in random forests, based on how frequently each variable is used as a split within the causal forest: variables that appear more often are thus more important for predicting treatment effect heterogeneity. This generates a ranking that serves as a proxy for classifying the key predictors of the observed heterogeneity.

I run the algorithm both with and without firms' variables to illustrate differences in the impor-

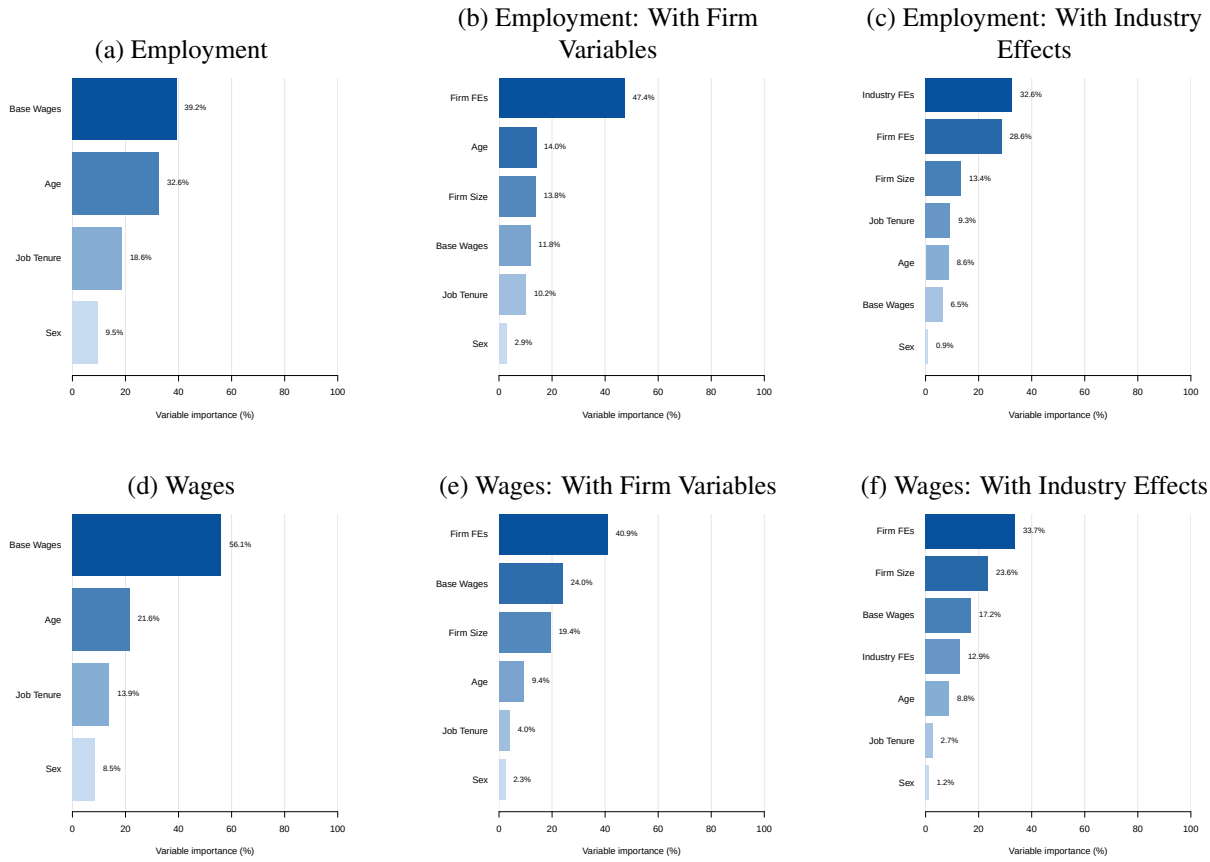
tance measure. I use the change in employment for each individual between 2018 and the average pre-shock period employment as the outcome. When excluding firms' variables, I find in Figure 8a that base wages are the most important predictor of heterogeneity in employment impacts, followed by age and job tenure. Notably, firm FEs are among the variables used most often in the splits once firms' variables are included, ahead of age and firm size. It is important to note that this measure does not indicate the sign or magnitude of each variable's effect on employment. Instead, it summarizes which variables the forest uses most frequently to distinguish heterogeneous treatment effects.

I then use the individual wage growth between 2018 and 2015 as the outcome. Without incorporating firms' variables, Figure 8d shows that base wages are the most important variable, followed by age and job tenure. Notably, when including firms' variables in the causal forest, Figure 8e shows that firm-specific pay premiums are among the variables used most often, ahead of base wages and firm size. Note that firm pay premiums and firm size are positively correlated, though not so strongly (correlation coefficient of 0.19). This additional evidence identifies firm-specific pay premiums and firm size, alongside worker characteristics, as key predictors for the heterogeneity in wage and employment effects. Firm FEs remain important predictors when including worker FEs, giving greater weight to early splits or discretizing continuous variables, although the rankings change across specifications. Online Appendix E presents these checks and discusses limitations of the importance rankings.

Lastly, since firm pay premiums correlate with industry type (Card et al., 2024) and immigrants concentrate in specific industries, I include the weighted average of firm FEs at the industry level in the algorithm.[60] Industry effects play a larger role for employment: they rank first at 32.6% of the importance measure, followed by firm FEs at 28.6% (see Figure 8c). For wages, firm FEs remain first at 33.7%, followed by firm size, while industry effects account for 12.9% (see Figure 8f).

[60] I group industries into 19 broader categories based on ISIC revision 4 to address potential misclassification in PILA's 4-digit codes.

Figure 8: Variable Importance for Employment and Wages in the Causal Forest



Note: The forests use 2018 wage changes relative to 2015 and employment changes relative to average employment in 2013–2015. Panels report depth-weighted variable importance using weights proportional to d^{-2} for selected splits through depth 4. Each forest contains 2,000 trees, clusters observations by FUA, and uses a minimum node size of 300. The sample is restricted to native employees aged 25 to 55. Industry effects are employment-weighted averages of firm FEs within 19 industry categories.

8 Conclusion

This paper joins a small but growing set of studies using administrative data to examine immigration-induced labor supply shocks in developing-country contexts, and is, to my knowledge, the first to do so for the Venezuelan exodus into Colombia, exploiting data that covers the entire universe of formal workers and firms. This administrative panel has advantages over prior survey-based studies of this setting. First, administrative panel data track workers over time, addressing compositional changes that typically arise in the standard regional-level analysis of immigration based on

cross-sectional surveys. Second, the matched employee-employer structure of the data can identify firm-level heterogeneity, which is crucial for understanding the mechanisms at play following immigration shocks in a context with informality and binding minimum wages.

My findings suggest that the arrival of Venezuelan immigrants has a negative impact on the employment and wages of individuals in the Colombian formal sector in the short term. However, this coefficient masks important heterogeneous responses. Minimum-wage workers are crowded out of the formal sector, whereas higher-wage workers are not displaced but instead experience lower wage growth. The negative effect on employment and wages is also concentrated in small formal firms. This result is consistent with the theoretical framework, which predicts that small firms that rely more heavily on informal labor will reduce formal employment and wages more significantly following an immigration shock. Beyond firm size, firm pay premiums are also a key predictor of the heterogeneity in immigration effects, as I find workers in low-paying firms experience a more negative impact on their earnings.

Given substantial heterogeneity across workers and firms, I use a data-driven approach to identify which variables best capture the variation in employment and wage effects. Throughout this analysis, firm pay premiums are consistently among the variables the algorithm points out as key for the heterogeneity, together with firm size in most cases. This additional evidence complements the paper's main results: research on the labor market impacts of immigration should not focus solely on workers' characteristics, as this yields an incomplete picture of the main sources of adjustment. After immigrants arrive, the focus should be on both *who* the worker is and the *type of firm* where the worker is employed.

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Online Appendix

Worker Responses to Immigration Across Firms: Evidence from Colombia

Lukas Delgado-Prieto

A Derivation of Model in Section 3

Throughout the derivations, I assume $0 < \varepsilon < 1$, $\rho < 1$, $\beta_I, \beta_F > 0$, and $\eta \geq 1$, with positive baseline employment of both worker types.

In this model, the indirect utility of worker i employed at firm j is:

$$v_{i,L_j} = \beta_L \ln w_{L_j} + a_{L_j} + \nu_{i,L_j}. \tag{A.1}$$

Under the assumption that ν_{i,L_j} follows a type I extreme value distribution for each of the workers' types $L \in \{I, F\}$ and that the number of firms J is sufficiently large, Card et al. (2018) shows that the firm-specific labor supply functions can be expressed as:

$$\ln I_j = \ln(\mathcal{I}\lambda_I) + \beta_I \ln w_{I_j} + a_{I_j}, \tag{A.2}$$

$$\ln F_j = \ln(\mathcal{F}\lambda_F) + \beta_F \ln w_{F_j} + a_{F_j}. \tag{A.3}$$

In these equations, the total number of informal workers in the market is $\mathcal{I}$ and of formal workers is $\mathcal{F}$, where λ_I and λ_F are constant parameters across firms.

Here, $\frac{d \ln L(w_{L_j})}{d \ln w_{L_j}} = \beta_L$ is the elasticity of labor supply to the firm with respect to its wage. Hence, as $\beta_L \to \infty$, the supply functions become perfectly elastic, and firms have no monopsony power to set wages below the marginal product of labor.[A.1] Adding the monopsony angle to the model is important for explaining equilibrium wage differentials across firms, consistent with my findings below with the AKM regression. However, my focus differs from that of Amior and Stuhler (2022), which examines the mediating role of monopsony power under immigration shocks. Instead, I explore how having firms that can hire both formal and informal labor generates different predictions about the impact of immigration shocks on workers, taking monopsony power as constant.

[A.1] I omit any market wage offered in an outside competitive sector as the comparative statics focus is on firm-level responses to immigration and not on market-level responses that have been thoroughly analyzed in the migration literature.

I then explain the derivations of the Equations in subsection 3. First, to derive the firm-specific optimal wages, I maximize the profit Equation (1) for each type of worker:[A.2]

$$\frac{d\pi_j}{dI_j} = 0 \Leftrightarrow w_{I_j} = \left(\frac{\beta_I}{1+\beta_I(1+\eta)}\right) D_j T_j^{\varepsilon} \varepsilon \alpha_I I_j^{\rho-1-\eta} (\alpha_I I_j^{\rho} + \alpha_F F_j^{\rho})^{\frac{\varepsilon-\rho}{\rho}}, \tag{A.4}$$

$$\frac{d\pi_j}{dF_j} = 0 \Leftrightarrow w_{F_j} = \left(\frac{\beta_F}{1+\beta_F}\right) D_j T_j^{\varepsilon} \varepsilon \alpha_F F_j^{\rho-1} (1+\tau_F)^{-1} (\alpha_I I_j^{\rho} + \alpha_F F_j^{\rho})^{\frac{\varepsilon-\rho}{\rho}}. \tag{A.5}$$

Here, workers' wages not only depend on their marginal productivity but also on the labor supply elasticities to the firm.[A.3] For clarity, I take logarithms of the wages in Equations (A.4) and (A.5):

$$\ln w_{I_j} = \ln\left(\frac{\beta_I}{1+\beta_I(1+\eta)}\right) + \ln(D_j T_j^{\varepsilon} \varepsilon \alpha_I) + (\rho - 1 - \eta)\ln I_j + \left(\frac{\varepsilon-\rho}{\rho}\right)\ln(\alpha_I I_j^{\rho} + \alpha_F F_j^{\rho}), \tag{A.6}$$

$$\ln w_{F_j} = \ln\left(\frac{\beta_F}{1+\beta_F}\right) + \ln(D_j T_j^{\varepsilon} \varepsilon \alpha_F) + (\rho - 1)\ln F_j - \ln(1+\tau_F) + \left(\frac{\varepsilon-\rho}{\rho}\right)\ln(\alpha_I I_j^{\rho} + \alpha_F F_j^{\rho}). \tag{A.7}$$

I first consider firms for which the minimum wage does not bind. Broadly, this model predicts that firms with higher productivity (T_j) or demand (D_j) will pay higher wages, holding amenities constant. I then study how firm-level wages respond to an immigration shock that shifts the aggregate informal labor supply outwards ($d\mathcal{I}$)[A.4]:

$$\frac{d\ln w_{I_j}}{d\mathcal{I}} \cdot \mathcal{I} = (\rho - 1 - \eta)\frac{d\ln I_j}{d\ln \mathcal{I}} + \left(\frac{\varepsilon-\rho}{\rho}\right)\frac{(\alpha_I \rho I_j^{\rho-1}\frac{dI_j}{d\mathcal{I}} + \alpha_F \rho F_j^{\rho-1}\frac{dF_j}{d\mathcal{I}})}{\alpha_I I_j^{\rho} + \alpha_F F_j^{\rho}} * \mathcal{I}, \tag{A.8}$$

$$\frac{d\ln w_{F_j}}{d\mathcal{I}} \cdot \mathcal{I} = (\rho - 1)\frac{d\ln F_j}{d\ln \mathcal{I}} + \left(\frac{\varepsilon-\rho}{\rho}\right)\frac{(\alpha_I \rho I_j^{\rho-1}\frac{dI_j}{d\mathcal{I}} + \alpha_F \rho F_j^{\rho-1}\frac{dF_j}{d\mathcal{I}})}{\alpha_I I_j^{\rho} + \alpha_F F_j^{\rho}} * \mathcal{I}. \tag{A.9}$$

Simplifying the last expressions and defining the derivatives as the elasticities, I find that:

$$\varepsilon_{w_{I_j},\mathcal{I}} = -(1+\eta-\rho)\varepsilon_{I_j,\mathcal{I}} + (\varepsilon-\rho)(s_{I_j}\varepsilon_{I_j,\mathcal{I}} + s_{F_j}\varepsilon_{F_j,\mathcal{I}}), \tag{A.10}$$

[A.2] In the derivations, I multiply by $\frac{w(L_j)}{w(L_j)}$ in the last term of FOCs to find the equations in the text.

[A.3] For formal workers, if $\beta_F = 9$ and there are no payroll taxes ($\tau_F = 0$), wages equal 90% of the marginal revenue product of labor.

[A.4] Assuming that the supply shock does not affect the firm-specific demand and the firm-specific amenities for each group of workers. Besides, the number of firms is sufficiently large that there are no strategic interactions among them.

$$\varepsilon_{w_{F_j},\mathcal{J}} = -(1-\rho)\varepsilon_{F_j,\mathcal{J}} + (\varepsilon-\rho)(s_{I_j}\varepsilon_{I_j,\mathcal{J}} + s_{F_j}\varepsilon_{F_j,\mathcal{J}}). \tag{A.11}$$

In these expressions, $s_{L_j} = \frac{\alpha_L L_j^\rho}{\alpha_I I_j^\rho + \alpha_F F_j^\rho}$ is the relative contribution of type of worker $L \in \{I,F\}$ to production. To further derive these elasticities, I use the changes in the firm-specific supply functions (A.2) and (A.3) after an immigration shock:

$$\varepsilon_{I_j,\mathcal{J}} = 1 + \beta_I \varepsilon_{w_{I_j},\mathcal{J}}, \tag{A.12}$$

$$\varepsilon_{F_j,\mathcal{J}} = \beta_F \varepsilon_{w_{F_j},\mathcal{J}}. \tag{A.13}$$

This yields a direct relationship between wages and employment as a function of the elasticities of supply to the firm.[A.5] Then, I replace Equations (A.12) and (A.13) into (A.10) and into (A.11):

$$\varepsilon_{w_{I_j},\mathcal{J}} = -(1+\eta-\rho)(1+\beta_I\varepsilon_{w_{I_j},\mathcal{J}}) + (\varepsilon-\rho)(s_{I_j}(1+\beta_I\varepsilon_{w_{I_j},\mathcal{J}}) + s_{F_j}\beta_F\varepsilon_{w_{F_j},\mathcal{J}}), \tag{A.14}$$

$$\varepsilon_{w_{F_j},\mathcal{J}} = -(1-\rho)\beta_F\varepsilon_{w_{F_j},\mathcal{J}} + (\varepsilon-\rho)(s_{I_j}(1+\beta_I\varepsilon_{w_{I_j},\mathcal{J}}) + s_{F_j}\beta_F\varepsilon_{w_{F_j},\mathcal{J}}). \tag{A.15}$$

Rearranging these expressions, I find that:

$$\varepsilon_{w_{I_j},\mathcal{J}} = \left(\frac{1}{\xi_{I_j}}\right)(-(1+\eta-\rho) + (\varepsilon-\rho)(s_{I_j} + s_{F_j}\beta_F\varepsilon_{w_{F_j},\mathcal{J}})), \tag{A.16}$$

$$\varepsilon_{w_{F_j},\mathcal{J}} = \left(\frac{1}{\xi_{F_j}}\right)(\varepsilon-\rho)s_{I_j}(1+\beta_I\varepsilon_{w_{I_j},\mathcal{J}}). \tag{A.17}$$

Here, I define $\xi_{I_j} = 1 + (1+\eta-\rho)\beta_I - (\varepsilon-\rho)s_{I_j}\beta_I$ and $\xi_{F_j} = 1 + (1-\rho)\beta_F - (\varepsilon-\rho)s_{F_j}\beta_F$. Then, replacing Equation (A.16) into (A.17) yields:

$$\varepsilon_{w_{F_j},\mathcal{J}} = \Omega_j s_{I_j}\beta_I(\varepsilon-\rho)\left(\frac{\xi_{I_j}}{\beta_I} - (1+\eta-\rho) + (\varepsilon-\rho)s_{I_j}\right). \tag{A.18}$$

Here, I define $\Omega_j = \frac{1}{\xi_{I_j}\xi_{F_j} - (\varepsilon-\rho)^2 s_{I_j}\beta_I s_{F_j}\beta_F}$. Under the parameter restrictions above, its denomi-

[A.5] Here, the total number of formal workers $\mathcal{F}$ in the market is held constant. Besides, in this partial equilibrium framework, the response of one firm does not have spillover effects on other firms.

nator can be written as

$$\Omega_j^{-1} = s_{F_j}\left[1+(1+\eta-\rho)\beta_I\right]\left[1+(1-\varepsilon)\beta_F\right] + s_{I_j}\left[1+(1+\eta-\varepsilon)\beta_I\right]\left[1+(1-\rho)\beta_F\right] > 0.$$

Thus, $\Omega_j > 0$. Last, I replace ξ_{I_j} inside of (A.18) to find the Equation (2) in the main text. Next, I plug Equation (2) inside Equation (A.16) to find that:

$$\varepsilon_{w_{I_j},\mathcal{J}} = \left(\frac{1}{\xi_{I_j}}\right)(-(1+\eta-\rho)+(\varepsilon-\rho)s_{I_j}(1+s_{F_j}\Omega_j(\varepsilon-\rho)\beta_F)). \tag{A.19}$$

In this case, the elasticity is going to be negative $\varepsilon_{w_{I_j},\mathcal{J}} < 0$.[A.6] Finally, after finding that informal wages always decrease with the informal labor supply shock, the last adjustment to analyze is what happens to informal employment within the firm. For that, I plug Equation (A.19) into Equation (A.12):

$$\varepsilon_{I_j,\mathcal{J}} = 1+\left(\frac{\beta_I}{\xi_{I_j}}\right)(-(1+\eta-\rho)+(\varepsilon-\rho)s_{I_j}(1+s_{F_j}\Omega_j(\varepsilon-\rho)\beta_F)). \tag{A.20}$$

After simplifying the previous expression, I find that:

$$\varepsilon_{I_j,\mathcal{J}} = \frac{1}{\xi_{I_j}}(1+(\varepsilon-\rho)^2 s_{I_j}\beta_I s_{F_j}\beta_F\Omega_j). \tag{A.21}$$

Thus, in this case, a positive aggregate informal shock always increases informal labor within the firm ($\varepsilon_{I_j,\mathcal{J}} > 0$), independent of whether formal and informal workers are close substitutes or not.

Case when the minimum wage binds for certain firms ($\overline{w}_{F_{Min}} \geq w_{F_j}$).

Among firms paying the minimum wage, it is useful to distinguish between two regimes. Some firms will be *demand-constrained* by the minimum wage, in the sense that formal employment is pinned down by equating $\text{MRPL}_{F_j} = (1+\tau_F)\overline{w}_{F_{Min}}$, while labor supply at the minimum wage is slack, meaning they do not hire all workers willing to work for that wage. Other firms will be *supply-constrained*, in which case formal employment is directly pinned down by the labor supply equation evaluated at the minimum wage and does not respond to an informal supply shock as the minimum wage is held fixed. In what follows, we focus on demand-constrained firms that pay the minimum wage and respond to the immigration shock.

Define the demand-constrained firms as j_m. In those firms, the formal wage response is muted,

[A.6] It suffices for $\varepsilon_{w_{I_j},\mathcal{J}} < 0$ that $1 \geq s_{I_j}\left(1+s_{F_j}\Omega_j(\varepsilon-\rho)\beta_F\right)$, which always holds when $\rho > \varepsilon$. If $\rho < \varepsilon$, the inequality still holds since $1+\eta-\rho > \varepsilon-\rho$.

$\varepsilon_{\overline{w}_{F_{Min}},\mathcal{J}} = 0$, so that the formal employment elasticity is equal to:

$$\varepsilon_{F_{jm},\mathcal{J}} = \frac{(\varepsilon-\rho)\, s_{I_{jm}}}{(1-\varepsilon)+(\varepsilon-\rho)\, s_{I_{jm}}}\, \varepsilon_{I_{jm},\mathcal{J}}. \tag{A.22}$$

Then, the informal wage elasticity for these firms can be derived from Equations A.10, A.12, and A.22, and it is equal to:

$$\begin{aligned}\varepsilon_{w_{I_{jm}},\mathcal{J}} = & -(1+\eta-\rho)\left(1+\beta_I \varepsilon_{w_{I_{jm}},\mathcal{J}}\right) \\ & +(\varepsilon-\rho)\left[s_{I_{jm}} + s_{F_{jm}} \frac{(\varepsilon-\rho) s_{I_{jm}}}{(1-\varepsilon)+(\varepsilon-\rho) s_{I_{jm}}}\right]\left(1+\beta_I \varepsilon_{w_{I_{jm}},\mathcal{J}}\right).\end{aligned} \tag{A.23}$$

This expression simplifies to:

$$\varepsilon_{w_{I_{jm}},\mathcal{J}} = \frac{-(1+\eta-\rho)+(\varepsilon-\rho) s_{I_{jm}}\left(1+s_{F_{jm}} \frac{(\varepsilon-\rho)}{(1-\varepsilon)+(\varepsilon-\rho) s_{I_{jm}}}\right)}{1+\beta_I\left((1+\eta-\rho)-(\varepsilon-\rho) s_{I_{jm}}\left(1+s_{F_{jm}} \frac{(\varepsilon-\rho)}{(1-\varepsilon)+(\varepsilon-\rho) s_{I_{jm}}}\right)\right)}. \tag{A.24}$$

Holding the model parameters and baseline input shares fixed, this negative elasticity is smaller in absolute value than that of firms for which the minimum wage does not bind. Formal wages remain fixed, while both formal and informal employment can adjust. From this expression, the informal employment elasticity is:

$$\begin{aligned}\varepsilon_{I_{jm},\mathcal{J}} &= 1+\beta_I \varepsilon_{w_{I_{jm}},\mathcal{J}} \\ &= \left[1+\beta_I\left((1+\eta-\rho)-(\varepsilon-\rho) s_{I_{jm}}\left(1+s_{F_{jm}} \frac{\varepsilon-\rho}{(1-\varepsilon)+(\varepsilon-\rho) s_{I_{jm}}}\right)\right)\right]^{-1}.\end{aligned} \tag{A.25}$$

This elasticity is again positive and larger than that of firms where the minimum wage does not bind. Finally, combining this result with equation (A.22), the formal employment elasticity for demand-constrained firms paying the minimum wage is:

$$\begin{aligned}\varepsilon_{F_{jm},\mathcal{J}} = & \frac{(\varepsilon-\rho) s_{I_{jm}}}{(1-\varepsilon)+(\varepsilon-\rho) s_{I_{jm}}} \\ & \times\left[1+\beta_I\left((1+\eta-\rho)-(\varepsilon-\rho) s_{I_{jm}}\left(1+s_{F_{jm}} \frac{\varepsilon-\rho}{(1-\varepsilon)+(\varepsilon-\rho) s_{I_{jm}}}\right)\right)\right]^{-1}.\end{aligned} \tag{A.26}$$

For firms that are supply constrained at the minimum wage, formal employment is pinned down by the labor supply function and does not respond to the immigration shock, so that $\varepsilon_{F_{jm},\mathcal{J}} = 0$.

B Construction of the Main Sample

The administrative records of the PILA are constructed at the contract level, as workers with more than one labor contract must pay contributions for each one. To transform to a worker-level dataset, I first drop contributions to the health system of type *N*, which present corrections to base income or changes in labor status. Second, I sum the labor income for workers with multiple labor contracts and retain the characteristics of the highest-paying job for each worker.

To construct the sample for the AKM estimation, I restrict it to six years before the immigration shock to capture more movements of workers between firms. This sample uses the years 2010 to 2015 for August. The total sample consists of 32,195,048 worker-year observations after eliminating workers with non-positive wages, with fewer than 30 employment days per month, restricting to employees between 20 and 60 years, and leaving the highest-wage job for workers with more than one contribution to the social security system.[B.1] Also, I exclude 3,931,843 additional workers because they do not belong to the largest connected set of firms and workers or appear only once in the estimation sample. Then, the nominal wages are transformed to real terms using the monthly CPI from DANE (with the base year 2018) and the logarithms of the final expression ($\ln w_{it}$). Table B.1 shows descriptive statistics by the seven quantiles of firm FEs, and Table B.2 shows the decomposition of the variance of wages $Var(\ln w_{it})$.

With this in mind, the definitions of the variables I use in the main analysis are as follows.

1. **Formal wages**. I use each worker's nominal contribution to the health system in August. I consider only positive contributions, as zero indicates workers on leave for reasons unrelated to wages or jobs. I focus on workers who reported 30 days of employment.

2. **Natives with formal employment**. I define natives as individuals who appear in the PILA with a national identity card. I take all the natives in the sample with a non-negative wage as employed.

3. **Firms**. I only include workers classified as employees in the firm-level data, then aggregate by the firm identifier.

[B.1] Around 5% of workers in the PILA have more than one contribution.

Table B.1: Descriptive Statistics by Firm FEs

	Average				
7 quantiles of $\hat{\psi}_j$	Employment	Male share	Age	Real wages (USD)	N
1	8	0.6	37.7	239.2	40,201
2	18	0.7	37.1	224.0	41,628
3	14	0.6	37.2	232.5	37,703
4	13	0.6	37.5	248.3	36,223
5	18	0.5	38.0	276.4	36,599
6	40	0.5	38.3	342.0	38,524
7	81	0.5	38.4	616.1	42,455

Note: This table reports descriptive statistics by quantile of firm fixed effects in the PILA. Real wages are deflated using the CPI from DANE for prices in 2018. I convert Colombian pesos to USD using 2020 exchange rates from the World Bank. Only workers who contribute as employees are taken into account. Source: PILA, August 2015.

Table B.2: Variance Decomposition Of $\ln w_{it}$

Component	Value	Share of $Var(\ln w_{it})$
$Var(\alpha_i)$	0.222	50.1%
$Var(\psi_{j(i)})$	0.069	15.7%
$2 * Cov(\alpha_i, \psi_{j(i)})$	0.095	21.6%
$Var(\ln w_{it})$	0.443	100%
$Corr(\alpha_i, \psi_{j(i)})$	0.384	

Note: This Table reports the variance decomposition of wages in the formal sector in Colombia using the leave-one-out connected set of workers and firms with the method proposed in Kline et al. (2020) with year FEs as the control variable. Source: PILA, August 2010–August 2015.

C Survey Evidence on the Informal Sector

Because informal workers are not observable in the administrative data, I cannot directly show the effects on the informal sector using PILA. To complement the worker-level evidence, I exploit the labor force survey (GEIH, by its acronym in Spanish) to construct measures of employment, wages, and the informality share at the department-year level for workers aged 18 to 64. I define formality based on contributions to the social security system, consistent with the PILA definition, and estimate the event study specification from Delgado-Prieto (2024):

$$Y_{dt} = \gamma_d + \gamma_t + \sum_{t=2011, t\neq 2015}^{2019} \beta_t T_t \cdot \Delta M_{d,2018} + u_{dt}, \tag{C.1}$$

where Y_{dt} is log employment, log wages, or the informality share in department d, and $\Delta M_{d,2018}$ is the time-invariant immigration shock instrumented with the distance to the nearest crossing

bridge with Venezuela and past Venezuelan settlements, as described in Delgado-Prieto (2024). For each outcome, I report the average of the pre-treatment coefficients (2011–2014) and the 2018 coefficient.

Table C.1 reports the results for natives. Informal wages decline by 1.9% for a 1 percentage point increase in $\Delta M_{d,2018}$, consistent with a labor supply shock that lowers the price of informal labor. As informal labor becomes cheaper, firms that employ both types of workers shift toward informal employment. Native formal employment declines significantly (-1.9%), while informal employment shows a smaller negative effect (-1.0%). Formal wages have a negative but insignificant coefficient.

Table C.1: Native Employment and Wages by Formality Status

	Formal employment	Informal employment	Formal wages	Informal wages
	(1)	(2)	(3)	(4)
β_{pre}	-0.163	0.536	0.632	-0.330
	(0.744)	(0.466)	(0.654)	(0.533)
β_{2018}	-1.894^{***}	-1.043^{*}	-0.499	-1.896^{***}
	(0.477)	(0.531)	(0.399)	(0.639)
N	216	216	216	216

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Regressions are estimated at the department level for 24 departments from 2011 to 2019. The outcomes in columns (1)–(2) are changes in log employment for native formal and informal workers, and in columns (3)–(4) are changes in log hourly wages, all relative to 2015. β_{pre} reports the average of the pre-treatment coefficients (2011–2014). The immigration shock $\Delta M_{d,2018}$ is instrumented with the distance to the nearest crossing bridge with Venezuela and past Venezuelan settlements as in Delgado-Prieto (2024). The sample is restricted to native workers aged 18 to 64 in urban areas. Standard errors are clustered at the department level. Coefficients report employment (in percent) and wages (in percent) for a one-percentage-point increase in the immigration share. Hourly wages are in real terms using the monthly CPI from DANE. Source: GEIH, 2011–2019.

D Heterogeneity by Worker and Firm Characteristics

Workers experience varying employment and wage effects based on their characteristics and the types of firms they worked for before immigrants arrived. To determine the subgroups most affected in a standard way, I restrict the regressions to the intersection of subgroups where earlier findings indicate more negative coefficients.

First, Table D.1a shows that for minimum wage earners in 2015, immigration reduces the probability of formal sector employment by 1.5 pp. For the medium age group, the impact is less negative at 1.2 pp, whereas for self-employed workers, it is more negative at 2.2 pp. When combining these three characteristics, there are 565,594 workers in the sample, for whom the

negative effect of the immigration shock on the probability of being a formal worker is 2.6 pp.

Table D.1b divides the sample into subgroups with the highest negative coefficients for native wages. It shows that for workers earning more than the minimum wage in 2015, migration reduces average wages by 0.7%. For workers in the smallest firms in 2015, the impact is more negative at 0.8%, while for workers in middle-paying firms in 2015, the estimate is also 0.8%. When combining these characteristics, there are 30,772 workers in the sample, and the effect on wages in 2018 is a 1.9% reduction for a 1 pp increase in the immigration shock. However, this analysis is subject to arbitrary sample restrictions and smaller sample sizes, which can lead to differential effects partly due to random variation or statistical noise. Therefore, in the next section, I propose a method to estimate heterogeneous immigration effects in a data-driven way.

Table D.1: Most Affected Native Workers: Employment and Wages

(a) Employment

	(1)	(2)	(3)	(4)	(5)
Prob. of Employment	-0.841***	-1.453***	-1.188***	-2.194***	-2.647***
	(0.192)	(0.231)	(0.227)	(0.327)	(0.388)
Sample restriction					
Minimum wage earners	✗	✓	✗	✗	✓
Median age (35 years or more)	✗	✗	✓	✗	✓
Self-employed	✗	✗	✗	✓	✓
N	6,706,035	2,205,814	3,915,188	1,103,384	565,594
Clusters	109	109	109	109	109

(b) Wages

	(1)	(2)	(3)	(4)	(5)
Wages	-0.600*	-0.711*	-0.827**	-0.804**	-1.908***
	(0.239)	(0.315)	(0.320)	(0.260)	(0.477)
Sample restriction					
Above minimum wage	✗	✓	✗	✗	✓
Small firm (1 to 19 workers)	✗	✗	✓	✗	✓
Middle-paying firm (quantile 4)	✗	✗	✗	✓	✓
N	4,090,973	2,639,040	643,346	195,647	30,772
Clusters	109	109	109	109	109

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: Reported coefficients are θ_{2018}. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. I estimate the equation separately by subgroups. For Panel A, the outcome variable is $e_{i,2018} - \sum_{t=2013}^{2015} e_{it}/3$ where e_{it} indicates formal sector employment. For Panel B, the outcome variable is $\frac{w_{i,2018}-w_{i,2015}}{w_{i,2015}}$ where w_{it} indicates wages in the formal sector. The sample is restricted to natives aged 25 to 55. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. Workers are observed in August of each year. Source: PILA, 2015–2018.

E Machine Learning

The benchmark specification that the algorithm uses is as follows:

$$\Delta Y_{i,l,2018} = TE(x_i)\Delta\hat{M}_{l,2018} + \Delta\varepsilon_{i,l,2018}, \tag{E.1}$$

x_i represents the value of the variables in X_i, and $TE(x_i)$ denotes the treatment effect. The outcome of interest, $\Delta Y_{i,l,2018}$, is the change in individual employment or wages in 2018 relative to the pre-shock period. $\hat{M}_{l,2018}$ is the predicted immigration shock after regressing the observed rate on the instruments. The worker and firm variables, including those constructed from the AKM model, are the partitioning variables f included in the vector X_f. All these features or variables

correspond to baseline characteristics in 2015 and include age, sex, job tenure, wages, firm FEs, worker FEs, and firm size. Baseline wages enter as a continuous predictor. Self-employed workers are excluded from this section because their firm-related characteristics are not comparable to those of employees. The machine learning algorithm proceeds as follows:

1. Start with 50% of the full sample P.[a] I use the remaining held-out sample for prediction after the algorithm is trained.

2. For each tree, take a random subsample, without replacement, of P and select candidate splitting variables from X_f.

3. For every possible value of one variable in X_f, the data is split into two partitions (say P_l and P_r) to run separate regressions of form (E.1) to estimate treatment effects for each partition. Choose the variable with its cutoff value that maximizes the difference in treatment effects using this formula:

$$(TE_l - TE_r)^2.^{b} \tag{E.2}$$

4. Observations with a value below or equal to the cutoff value are placed into a new left node, and observations with a value above are placed into a new right node of the decision tree.

5. Recursively split the resulting nodes until no admissible split improves the splitting criterion, using a target minimum node size of 300. Splits must also satisfy a balance constraint that prevents either child node from containing less than 5% of the parent node's information content.

6. To predict $TE(x_i)$ for a held-out observation i, the algorithm constructs weights from the training observations that fall into the same terminal nodes as i. It then estimates a local treatment effect using these forest weights. For OOB predictions within the training sample, only trees that did not use observation i are included.

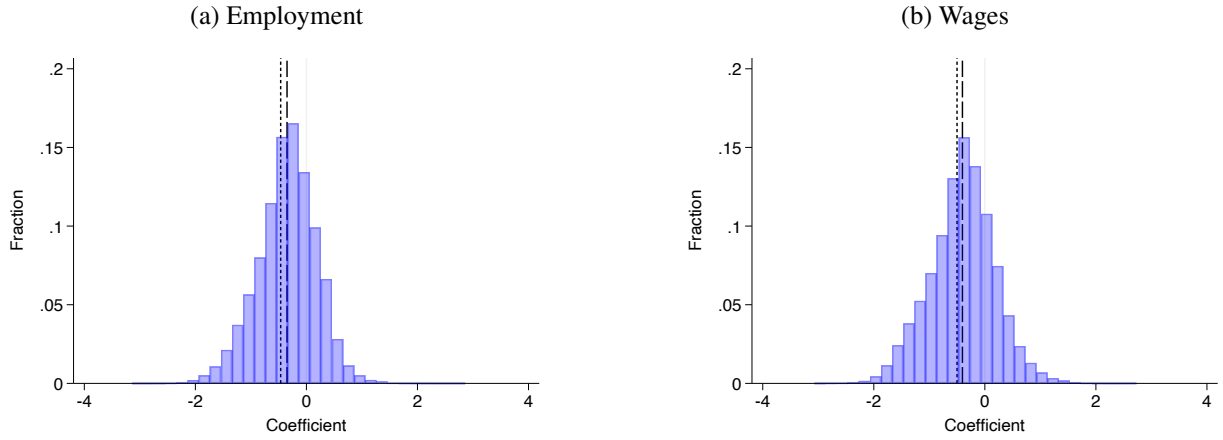
[a] The 50% threshold is selected due to computational burden. Within each tree, honesty uses separate subsamples to choose splits and estimate effects.

[b] There are penalties in the algorithm for the imbalance of the splits. For instance, the squared difference criterion can include an additional term $\frac{n_l n_r}{N^2}$ to adjust for more balanced splits (n_l and n_r refer to the sample size of each partition, and total subsample refers to N).

Construction and Distribution of CATEs. The first output of this procedure is shown in Online Appendix Figures E.1a and E.1b. These histograms show the predicted treatment effects on wage and employment outcomes, as estimated by the trained causal forest. The treatment effects shown here are predictions for the held-out sample, which is not used to train the forest. To estimate the conditional average treatment effect (CATE), each held-out observation is assigned to a terminal node of each tree based on its characteristics, and the algorithm uses the resulting forest weights to

estimate $TE(x_i)$. In the histograms, the long dashed line represents the average CATEs, while the short dashed line is the average treatment effect from the standard regression of Equation (4). For both outcomes, the average coefficient from the causal forest aligns with the standard regression, reflecting the accuracy of the average prediction.

Figure E.1: Histogram of Conditional Average Treatment Effects for Formal Employment and Formal Wages in the Causal Forest

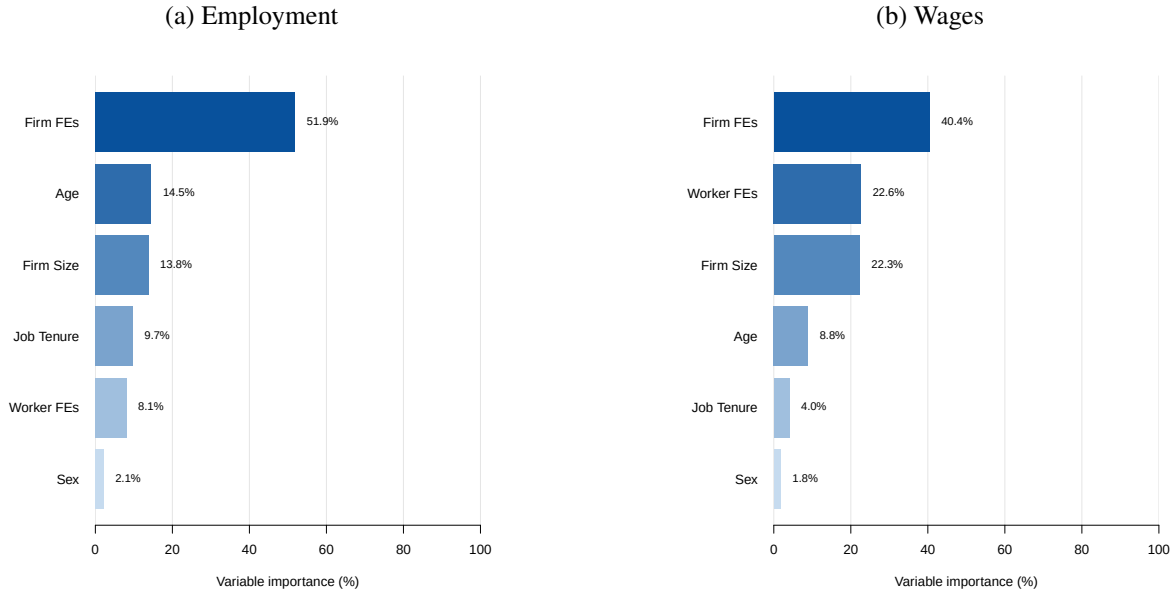


Note: The forests use 2018 wage changes relative to 2015 and employment changes relative to average employment in 2013–2015. The short dashed line refers to the coefficient from the benchmark specification, and the long dashed line refers to the average predicted treatment effects that are estimated with the trained causal forest using the held-out sample. The number of trees is 2,000. The sample is restricted to natives aged 25 to 55. I use clusters at the FUA level for the causal forest. The causal forest uses 50% of the main sample. The minimum node size is 300.

Robustness Checks of Variable Importance. First, since base wages are a function of the unobserved firm and worker FEs, I include the constructed worker FEs $\hat{\alpha}_i$ into the algorithm instead of base wages. This adjustment reduces the sample size, as every worker must be observed more than once. After including worker FEs, firm FEs and firm size remain among the variables most often used to predict heterogeneity (see Online Appendix Figures E.2a and E.2b). Second, early tree splits may not receive sufficiently larger weights than later ones, so I use depth weights proportional to d^{-4} instead of d^{-2}, where d denotes split depth, to give earlier splits greater relative weight. Firm FEs remain the most important variable for both outcomes (see Online Appendix Figures E.3a and E.3b). Third, because variables with fewer possible values can appear mechanically less important (Strobl et al., 2007), I discretize continuous variables into six or seven categories. With discretized variables, worker FEs rank first for wages, followed by firm FEs and firm size, while firm FEs rank first for employment (see Online Appendix Figures E.4a and E.4b). Nevertheless, exploiting the full range of variable values allows the algorithm to capture non-linearities and interactions more effectively than arbitrary categorization. Another concern is that the algorithm

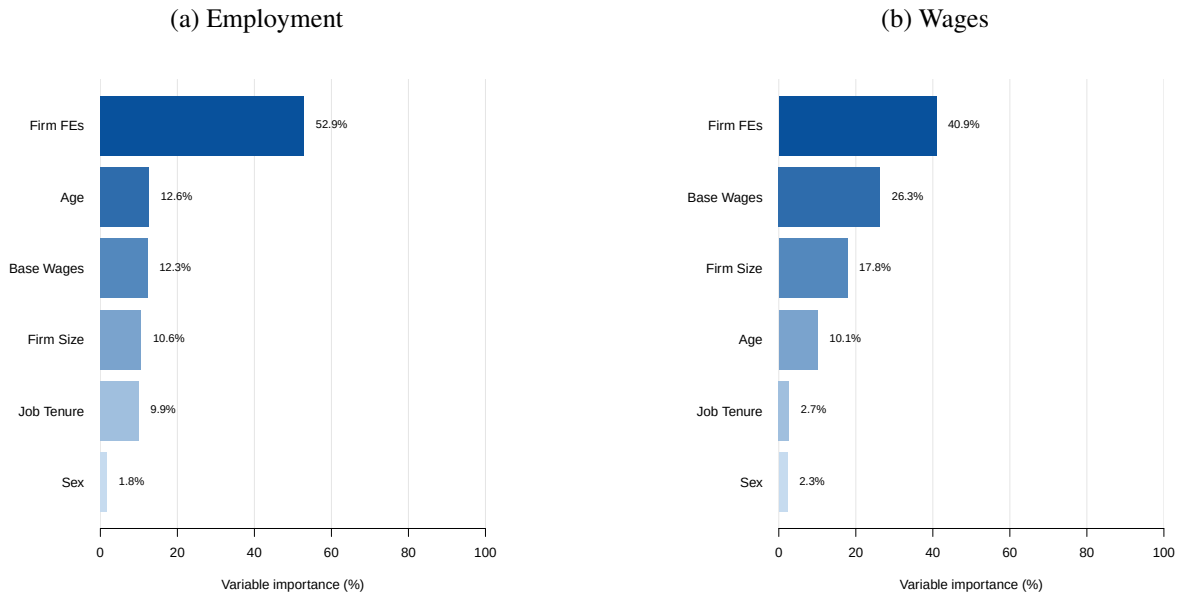
maximizes differences in treatment effects without testing pre-trends within subgroups, assuming strict exogeneity. Online Appendix F reports pre-trend checks across worker and firm subgroups, which are broadly flat across most subgroups, although not all. Fourth, although recent work proposes hypothesis testing for variable importance in random forests (Hapfelmeier et al., 2023), these methods are not yet available for causal forests and are computationally infeasible in settings with millions of observations.

Figure E.2: Variable Importance for Employment and Wages in Causal Forest With Worker and Firm FEs



Note: The forests use 2018 wage changes relative to 2015 and employment changes relative to average employment in 2013–2015. Variable importance is a simple weighted sum of how often the feature f appears in the splits across the leaves of each tree in the forest. For this exercise, the forest comprises 2,000 trees. The analysis sample is restricted to native employees aged 25 to 55. The importance measure uses weights proportional to d^{-2}, where d denotes split depth, and includes splits up to depth 4. The estimation is performed using clusters of FUAs in the causal forest. Additionally, the minimum node size for splitting is set to 300.

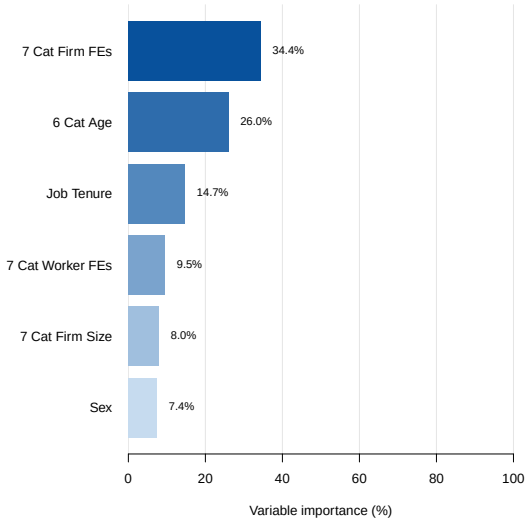
Figure E.3: Variable Importance for Employment and Wages in the Causal Forest With Stronger Decay Exponent



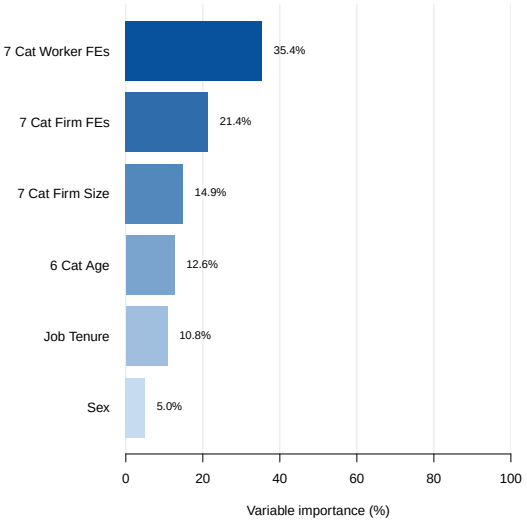
Note: The forests use 2018 wage changes relative to 2015 and employment changes relative to average employment in 2013–2015. Variable importance is a simple weighted sum of how often the feature f appears in the splits across the leaves of each tree in the forest. For this exercise, the forest comprises 2,000 trees. The analysis sample is restricted to native employees aged 25 to 55. The importance measure uses weights proportional to d^{-4}, where d denotes split depth, and includes splits up to depth 4. The estimation is performed using clusters of FUAs in the causal forest. Additionally, the minimum node size for splitting is set to 300.

Figure E.4: Variable Importance for Employment and Wages in the Causal Forest for Categories

(a) Employment



(b) Wages



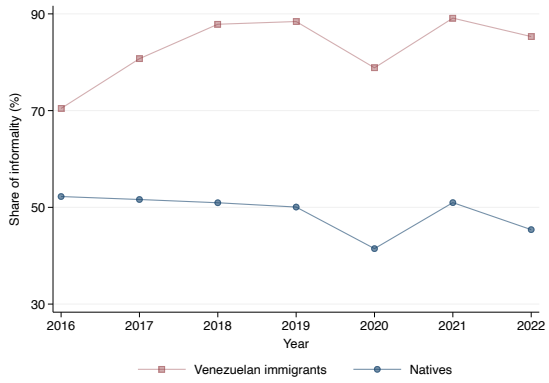
Note: The forests use 2018 wage changes relative to 2015 and employment changes relative to average employment in 2013–2015. Variable importance is a simple weighted sum of how often the feature f appears in the splits across the leaves of each tree in the forest. For this exercise, the forest comprises 2,000 trees. The analysis sample is restricted to native employees aged 25 to 55. The importance measure uses weights proportional to d^{-2}, where d denotes split depth, and includes splits up to depth 4. The estimation is performed using clusters of FUAs in the causal forest. Additionally, the minimum node size for splitting is set to 300.

F Supplementary Results

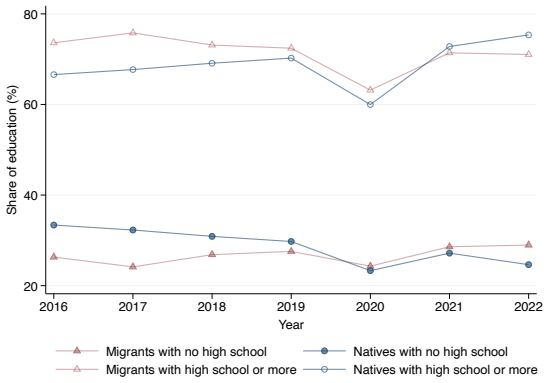
F.1 Figures

Figure F.1: Descriptive Statistics of Immigrants and Natives

(a) Informality

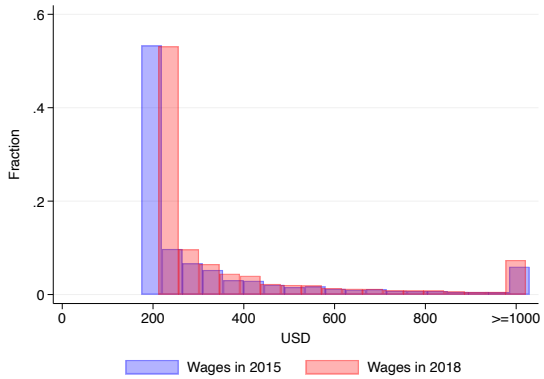


(b) Education



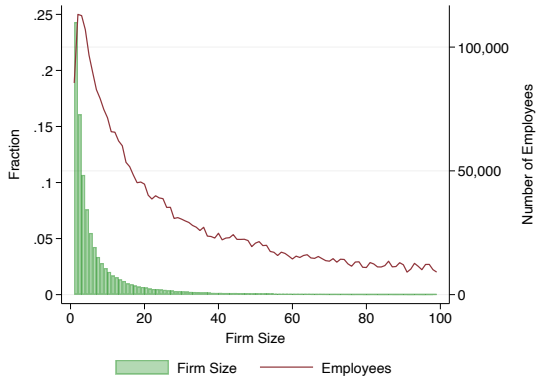
Note: I define informality based on contributions to the social security system and define the share over the employed population. For education, the share is over all the population. I aggregate information with national survey weights. Source: GEIH, 2016–2022.

Figure F.2: Histogram of Wages by Years



Note: The sample is restricted to native workers between 18 and 64 years with full employment days in the month and positive wages. Wages are in nominal terms. I convert Colombian pesos to USD using 2020 exchange rates from the World Bank. The chosen bin width is 45. Source: PILA, 2015–2018.

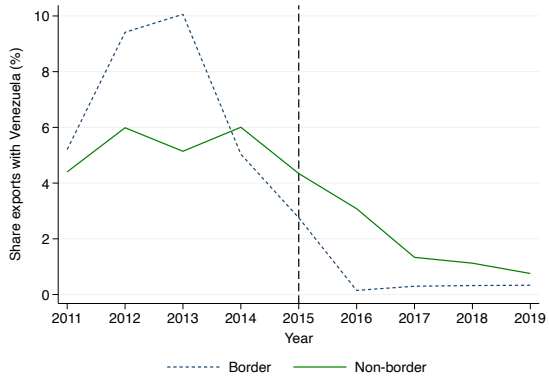
Figure F.3: Firm Size Distribution and Total Employees



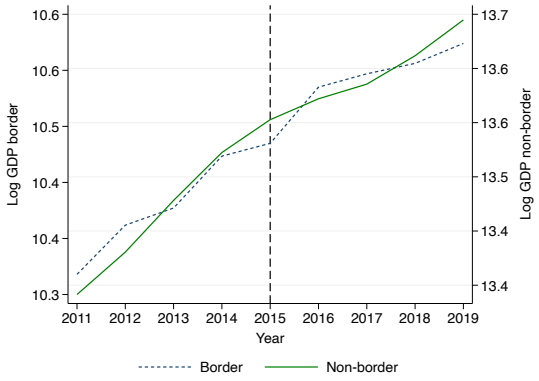
Note: The upper bound of firm size is restricted to 100 workers for the figure. The chosen bin width is 1. Only workers who contribute as employees are taken into account. Source: PILA, August 2015.

Figure F.4: Evolution of Trade and GDP for Border and Non-Border Departments

(a) Share Exports With Venezuela

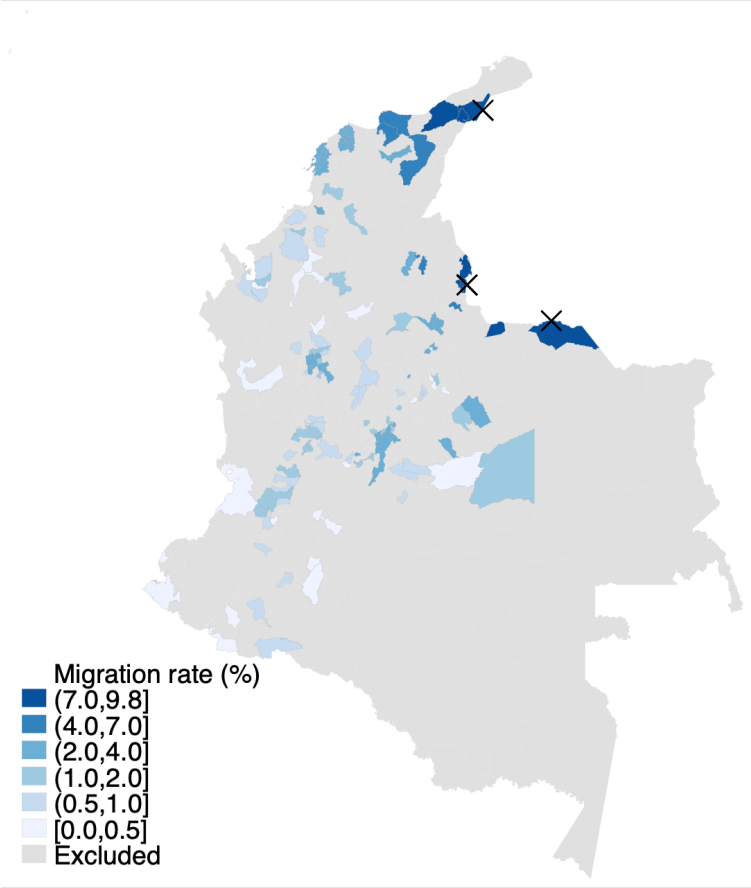


(b) Log GDP



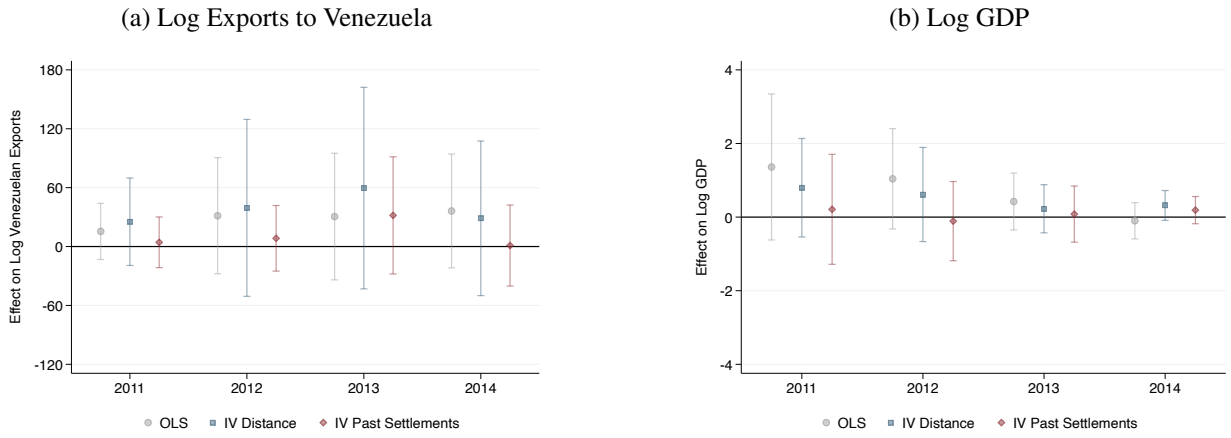
Note: Border departments are *Norte de Santander, La Guajira, and César*. Non-border departments are the rest. Source: Panel (a) *Exportaciones*–DANE, 2011–2019. Panel (b) DANE-*Cuentas Nacionales*, 2011–2019.

Figure F.5: Map of FUAs With the Immigration Shock $\Delta M_{l,2018}$



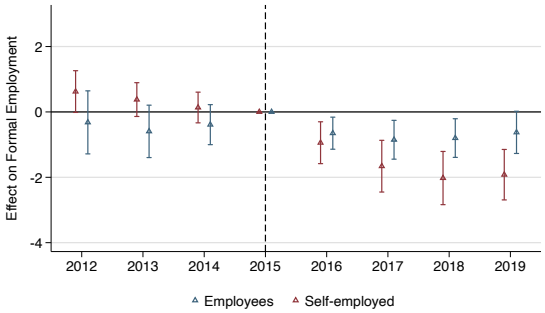
Note: The X represents the main three crossing bridges with Venezuela. The distance instrument is according to the nearest crossing bridge. Source: CNPV, 2018.

Figure F.6: Pre-Treatment Estimates on Venezuelan Exports and GDP at the Department-Level



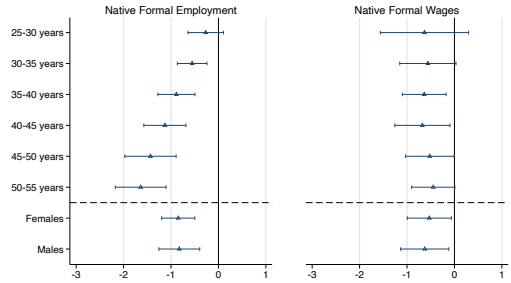
Note: The coefficients come from a department-level regression on 24 departments, based on the migration shock and instruments used in Delgado-Prieto (2024), and they are relative to the outcome in 2015. Coefficients report percentage changes for a one-percentage-point increase in the immigration share. Source: Panel (a) *Exportaciones*-DANE, 2011–2019. Panel (b) DANE-*Cuentas Nacionales*, 2011–2019.

Figure F.7: Event Study Estimates on Employment by Job Type



Note: I estimate Equation (4) separately by year and type of job. The sample is restricted to natives aged 25 to 55. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA 2012–2019.

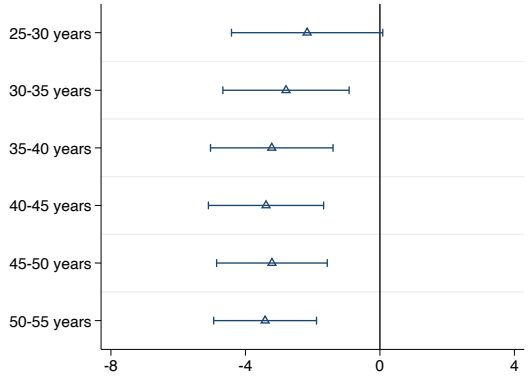
Figure F.8: Employment and Wage Estimates by Age Group



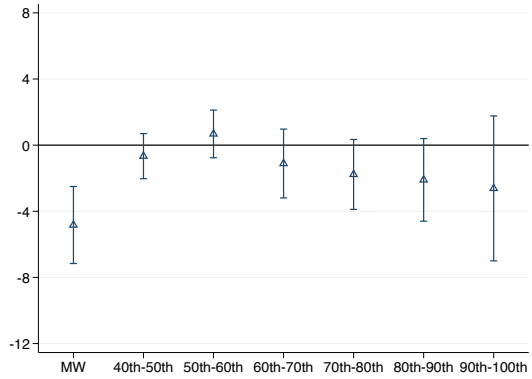
Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to natives aged 25 to 55. The dependent variables are employment and wages relative to the base period. I use as controls the interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019.

Figure F.9: Earnings Estimates by Different Worker and Firm Characteristics

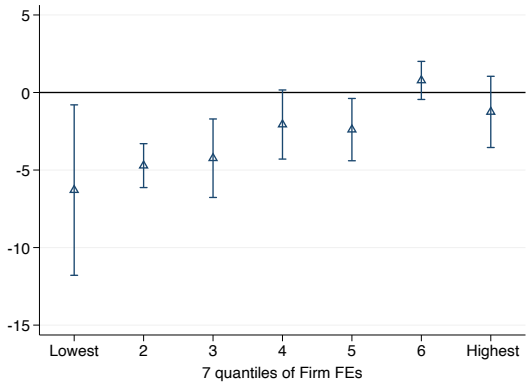
(a) By Age Groups



(b) By Wage Categories

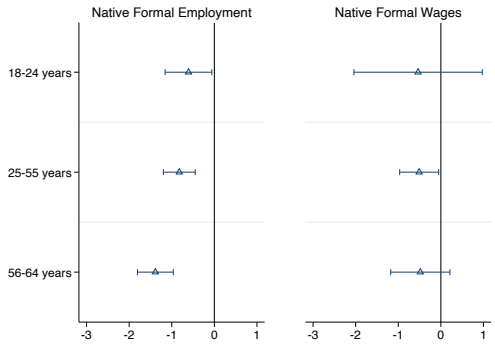


(c) By Pay Premiums



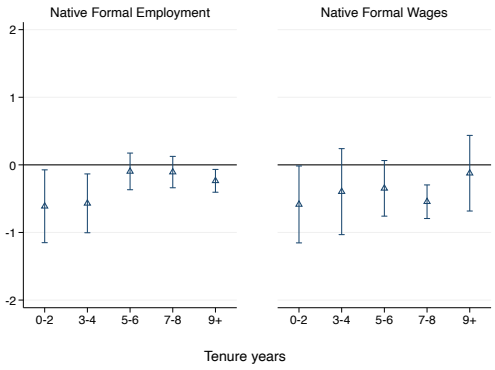
Note: The estimates cover cumulative formal earnings in 2016–2018, normalized by earnings in 2015. I estimate Equation (4) separately by subgroups. The sample is restricted to natives aged 25 to 55. The dependent variable is cumulative earnings in the post-treatment period. I use as controls the interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence intervals. Coefficients report earnings (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019.

Figure F.10: Estimates by Extended Age Categories



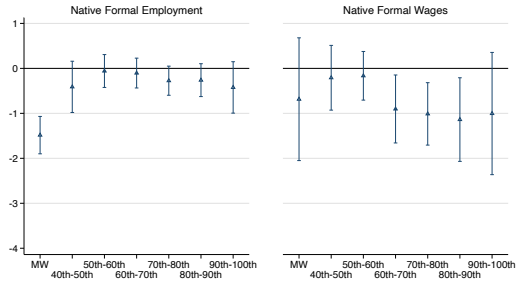
Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to natives between 18 and 64 years old. Dependent variables are employment relative to the pre-shock period and wages relative to the base period. I use as controls sex with a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019.

Figure F.11: Estimates by Job Tenure



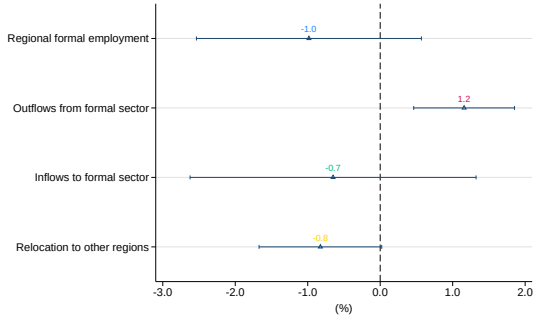
Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to native employees between 25 and 55 years old. Dependent variables are employment and wages relative to the base period. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. I constructed pre-shock job tenure from 2007 to 2015. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019.

Figure F.12: Estimates by Individual Wage Including a Labor Demand Control



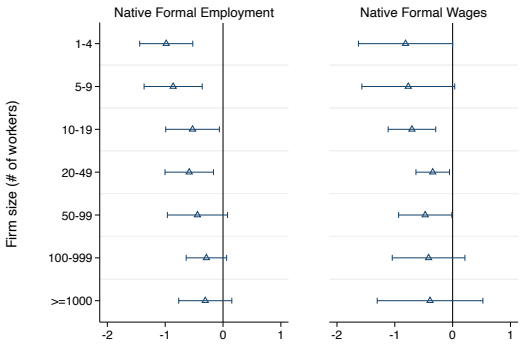
Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups including a Bartik-type control. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The control is the 2015–2018 log change in predicted local employment, $\ln \widehat{E}_{l,2018} - \ln \widehat{E}_{l,2015}$, where $\widehat{E}_{l,t} = \sum_i (E_{il,2015}/E_{i,2015}) E_{i,t}$. Here, $E_{il,2015}$ is baseline employment in industry i in FUA l, and $E_{i,t}$ is national employment in that industry in year t. I construct this measure using two-digit industries based on the ISIC revision 4 classification. The sample is restricted to natives aged 25 to 55. The dependent variables are employment relative to the pre-shock period and wages relative to the base period. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019 .

Figure F.13: Decomposition of Formal Employment Effects



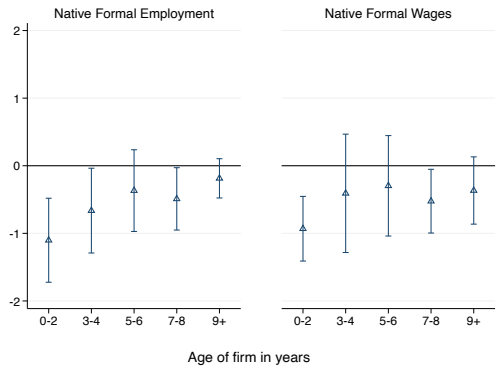
Note: Reported coefficients are θ_{2018}, measuring changes in 2018 relative to 2015. Regressions are estimated at the regional level for 109 FUAs weighted by their formal employment in 2015. 95% confidence interval. The sample is not restricted by age groups. Regional formal employment is decomposed into outflows from formal employment in that region, inflows from non-employment or the informal sector, employed people in other regions, and relocation of formal workers to other regions. Source: PILA, 2015–2018.

Figure F.14: Estimates by Firm Size Including Industry Fixed Effects



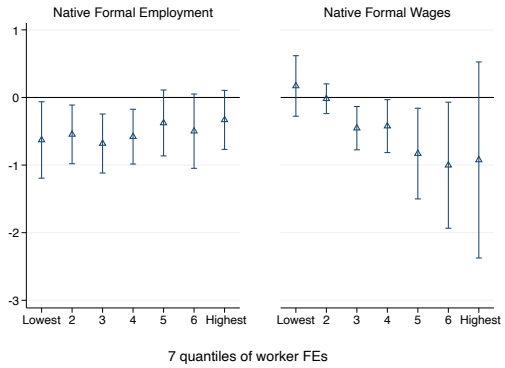
Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. I include 15 industry codes according to ISIC revision 4 classification. The sample is restricted to native employees between 25 and 55 years old. The dependent variables are employment relative to the pre-shock period and wages relative to the base period. I use as controls the interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Workers are observed in August of each year. Coefficients report employment (in percentage points) and for wages (in percent) for a one-percentage-point increase in the immigration share. Source: PILA, 2013–2019.

Figure F.15: Estimates by Age of Firm



Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to native employees between 25 and 55 years old. Dependent variables are employment relative to the pre-shock period and wages relative to the base period. Firm age is measured by the number of years in which the firm is observed in PILA, allowing for gaps. I use as controls the interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence intervals. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. Source: PILA, 2013–2019.

Figure F.16: Estimates by Quantiles of Worker FEs

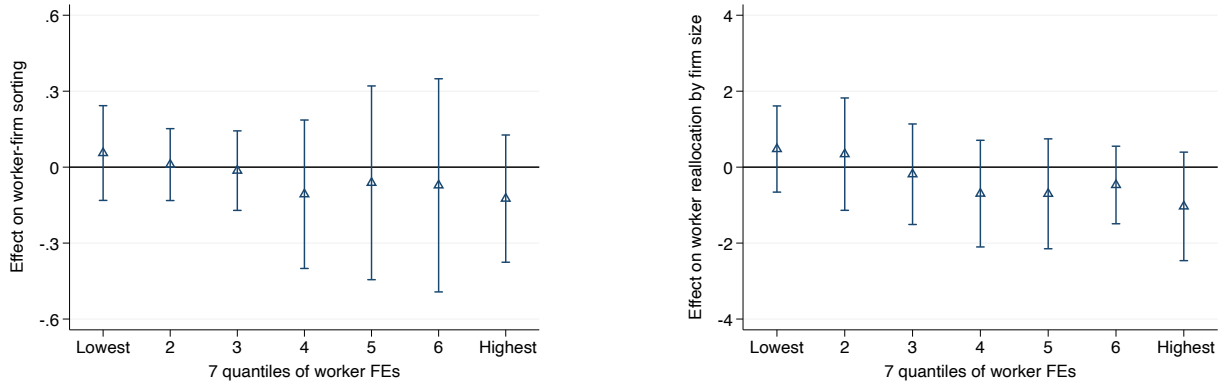


Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to native employees between 25 and 55 years old who appear more than once in PILA. Dependent variables are employment relative to the pre-shock period and wages relative to the base period. I compute Worker FEs in the first stage using the standard AKM framework, with age squared and its cubic as time-varying controls, for the period 2010-2015. I use as controls in the second stage interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. Coefficients report employment (in percentage points) and for wages (in percent) for a one-percentage-point increase in the immigration share. Workers are observed in August of each year. 95% confidence interval. Source: PILA, 2013–2019.

Figure F.17: Reallocation Estimates by Quantiles of Worker FEs

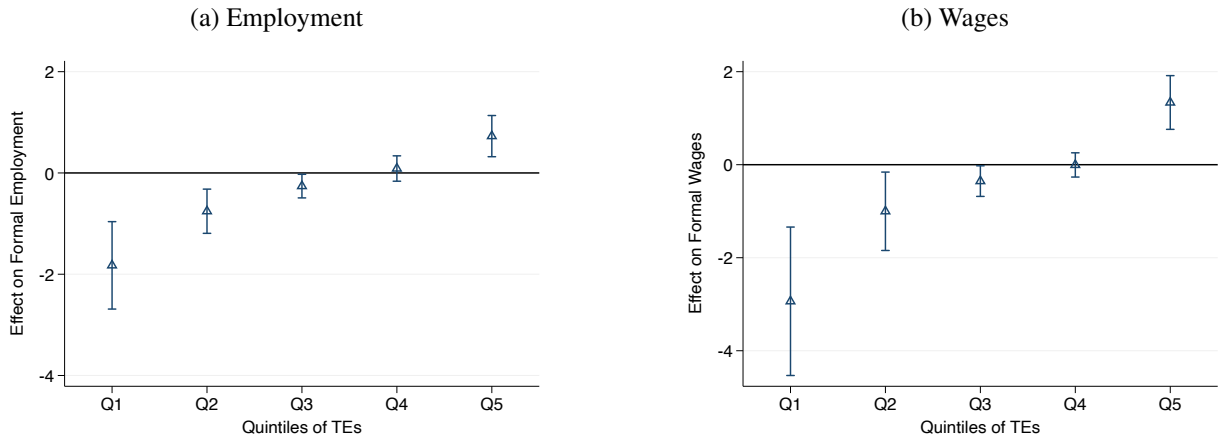
(a) By Pay Premiums

(b) By Firm Size



Note: Reported coefficients are θ_{2018}. The sample is restricted to natives aged 25 to 55. The dependent variable in (a) is the change in $\hat{\psi}_{i,\{j=2018\}} - \hat{\psi}_{i,\{j=2015\}}$ and in (b) is the change in the categories of firm size in 2018 relative to 2015, both measured in the base period. I use as controls interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. 95% confidence interval. Workers are observed in August of each year. Source: PILA, 2013–2019.

Figure F.18: Quintiles of CATEs for Employment and Wages in the Causal Forest



Note: Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The quintiles of treatment effects are constructed using the predicted conditional average treatment effects from the trained causal forest. The reported θ_{2018} coefficients come from separate regressions of Equation (4). The sample is restricted to natives aged 25 to 55. I use clusters at the FUA level for the causal forest. Standard errors are clustered at the level of 109 geographic units. 95% confidence intervals. The causal forest uses 50% of the main sample. Coefficients report employment (in percentage points) and wages (in percent) for a one-percentage-point increase in the immigration share.

F.2 Tables

Table F.1: Number of Observations by FUA I

	Observations	Percent
1. Bogotá	2,367,843	(33.2)
2. Medellín	983,096	(13.8)
3. Cali	593,447	(8.3)
4. Barranquilla	341,211	(4.8)
5. Cartagena	205,150	(2.9)
6. Bucaramanga	273,090	(3.8)
7. Cúcuta	110,123	(1.5)
8. Pereira	140,791	(2.0)
9. Ibagué	100,823	(1.4)
10. Manizales	103,401	(1.5)
11. Santa Marta	84,705	(1.2)
12. Pasto	70,170	(1.0)
13. Armenia	71,314	(1.0)
14. Villavicencio	106,493	(1.5)
15. Montería	71,007	(1.0)
16. Valledupar	76,072	(1.1)
17. Buenaventura	24,514	(0.3)
18. Neiva	71,376	(1.0)
19. Palmira	41,687	(0.6)
20. Popayán	62,422	(0.9)
21. Sincelejo	39,859	(0.6)
22. Barrancabermeja	35,095	(0.5)
23. Tuluá	25,123	(0.4)
24. Tunja	52,987	(0.7)
25. Riohacha	31,134	(0.4)
26. San Andres de Tumaco	7,960	(0.1)
27. Florencia	19,704	(0.3)
28. Apartadó	26,268	(0.4)
29. Girardot	14,920	(0.2)
30. Cartago	17,006	(0.2)
31. Maicao	6,263	(0.1)
32. Magangué	5,327	(0.1)
33. Sogamoso	18,220	(0.3)
34. Buga	21,072	(0.3)
35. Ipiales	8,754	(0.1)
36. Quibdó	15,687	(0.2)
37. Fusagasugá	12,899	(0.2)
38. Facatativá	18,796	(0.3)
39. Duitama	18,427	(0.3)
40. Yopal	43,279	(0.6)
41. Ciénaga	4,701	(0.1)
42. Zipaquirá	12,908	(0.2)
43. Rionegro	29,601	(0.4)
44. Ocaña	8,966	(0.1)
45. La Dorada	8,563	(0.1)
46. Caucasia	7,372	(0.1)
47. Sabanalarga	2,434	(0.0)
48. Aguachica	9,748	(0.1)
49. Espinal	6,439	(0.1)
50. Arauca	11,726	(0.2)
51. Santa Rosa de Cabal	4,887	(0.1)
52. El Carmen de Bolívar	1,411	(0.0)
53. Fundación	3,881	(0.1)
Continues in Table F.2		
No FUA assigned	417,188	(5.9)
Total	7,123,223	(100)

Note: This Table reports the number of workers from the PILA by FUAs 1 to 53. Percentages are relative to the 7,123,223 workers before the geographic sample restriction. The name represents the main city of FUA, but often they aggregate multiple municipalities according to Sanchez-Serra (2016). The sample is restricted to natives aged 25 to 55. Workers are observed in August of each year. Source: PILA, 2015.

Table F.2: Number of Observations by FUA II

	Observations	Percent
54. Acacías	12,472	(0.2)
55. Madrid	8,922	(0.1)
56. La Ceja	8,662	(0.1)
57. Santander de Quilichao	8,505	(0.1)
58. San Gil	8,268	(0.1)
59. Mocoa	7,974	(0.1)
60. Pitalito	7,852	(0.1)
61. Albania	7,020	(0.1)
62. Tocancipá	7,007	(0.1)
63. Los Patios	6,137	(0.1)
64. Montelíbano	6,083	(0.1)
65. Turbo	5,830	(0.1)
66. Granada	5,298	(0.1)
67. El Carmen de Viboral	5,047	(0.1)
68. Chinchiná	4,903	(0.1)
69. Puerto Boyacá	4,761	(0.1)
70. Guarne	4,697	(0.1)
71. Zarzal	4,584	(0.1)
72. Puerto Asís	4,568	(0.1)
73. Chiquinquirá	4,526	(0.1)
74. Villa de San Diego de Ubaté	4,522	(0.1)
75. Garzón	4,454	(0.1)
76. Santa Rosa de Osos	4,406	(0.1)
77. Puerto Gaitán	4,380	(0.1)
78. Pamplona	4,348	(0.1)
79. Puerto Tejada	4,279	(0.1)
80. Caloto	4,136	(0.1)
81. Segovia	4,016	(0.1)
82. Puerto Berrío	3,989	(0.1)
83. Lorica	3,875	(0.1)
84. Sopó	3,832	(0.1)
85. Aguazul	3,627	(0.1)
86. Santa Fé de Antioquia	3,589	(0.1)
87. Cereté	3,526	(0.0)
88. Puerto López	3,412	(0.0)
89. Pradera	3,388	(0.0)
90. La Cruz	3,387	(0.0)
91. La Virginia	3,375	(0.0)
92. San Pedro de los Milagros	3,170	(0.0)
93. Tenjo	3,166	(0.0)
94. Villanueva	3,136	(0.0)
95. Sahagún	3,126	(0.0)
96. Melgar	3,099	(0.0)
97. Barbosa, Santander	3,042	(0.0)
98. Socorro	3,026	(0.0)
99. Carepa	2,999	(0.0)
100. Planeta Rica	2,893	(0.0)
101. Chigorodó	2,880	(0.0)
102. Yarumal	2,874	(0.0)
103. Paipa	2,873	(0.0)
104. Samacá	2,782	(0.0)
105. Barbosa, Antioquia	2,781	(0.0)
106. Saravena	2,730	(0.0)
107. El Cerrito	2,597	(0.0)
108. Amagá	2,534	(0.0)
109. Villeta	2,518	(0.0)

Note: This Table reports the number of workers from the PILA by FUAs 54 to 109. Percentages are relative to the 7,123,223 workers before the geographic sample restriction. The name represents the main municipality. The sample is restricted to natives aged 25 to 55. Workers are observed in August of each year. Source: PILA, 2015.

Table F.3: Descriptive Statistics for Natives and Immigrants in the Formal Sector

(a) Colombians

	Age		Male share		Real Wages (USD)		
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	N
2013	37.0	10.8	0.56	0.496	423.3	506.2	7,335,933
2015	37.2	11.1	0.56	0.497	424.2	493.1	8,391,804
2017	37.8	11.4	0.55	0.497	415.7	482.8	8,064,238
2019	38.2	11.7	0.55	0.498	440.0	510.2	8,363,166

(b) Venezuelans With PEP

	Age		Male share		Real Wages (USD)		
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	N
2018	30.8	7.8	0.68	0.467	245.4	99.8	12,842
2019	31.8	8.1	0.67	0.472	250.9	99.4	42,752

(c) Other Foreigners

	Age		Male share		Real Wages (USD)		
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	N
2013	40.2	10	0.61	0.487	1,315.8	1,542.6	20,977
2015	39.5	10.2	0.64	0.481	1,275.7	1,472.3	27,729
2017	39.3	10.3	0.63	0.483	1,061.8	1,338.9	31,552
2019	39.6	10.4	0.58	0.494	999.3	1,304.5	39,703

Note: This table reports the descriptive statistics for Colombians, foreigners, and Venezuelans with PEP between 18 and 64 years of age. Only workers with full days of employment recorded in PILA and a positive health contribution are considered for wages and the number of observations. I only observe Venezuelans with PEP since 2018. The real wages are deflated using the Consumer Price Index (CPI) from DANE for prices in 2018. I convert Colombian pesos to USD using 2020 exchange rates from the World Bank. For self-employed workers, observed wages in PILA correspond to 40% or more of their actual wages by law, with the minimum wage as a lower bound. Source: PILA, 2013–2019, August.

Table F.4: Descriptive Statistics by Firm Size

		Average			
Firm size (# of workers)	Employment	Male share	Age	Real Wages	Firms
1-4	2	0.56	39.6	271.9	206,456
5-9	7	0.60	37.5	304.2	64,347
10-19	13	0.61	37.2	329.2	42,207
20-49	30	0.63	36.9	360.9	28,625
50-99	69	0.65	36.7	394.8	10,032
100-999	259	0.63	36.9	443.1	10,107
1000 and more	2677	0.58	36.0	530.8	859

Note: This Table reports the descriptive statistics for seven groups of firm size. I deflate real wages using the CPI from DANE for 2018 prices. Then, I transform Colombian pesos to USD using 2020 exchange rates from the World Bank. I only consider employees when constructing firm sizes. Source: PILA, 2015-August.

Table F.5: First Stage: The Inflow of Venezuelan Immigrants and the Two Instruments

	(1)	(2)	(3)
	$\Delta M_{l,2018}$	$\Delta M_{l,2018}$	$\Delta M_{l,2018}$
Distance (/100)	-1.992***		-1.455***
	(0.272)		(0.350)
Distance (/100) squared	0.151***		0.107***
	(0.024)		(0.029)
Past Settlements		0.703***	0.280*
		(0.160)	(0.130)
Constant	6.762***	1.040***	5.184***
	(0.715)	(0.149)	(1.000)
R^2	0.583	0.450	0.618
F	34.53	19.37	23.68
N	109	109	109

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the coefficient of the first-stage of the immigration shock $\Delta M_{l,2018} * 100$ with distance and distance squared to the nearest crossing bridge and past settlements as explanatory variables.

Table F.6: Robustness Checks for the Main Outcomes

	Employment	Wages
Baseline Specification (Both Instruments)	-0.841***	-0.600*
	(0.192)	(0.239)
N	6,706,035	4,090,973
Distance Instrument Only	-0.901**	-0.667*
	(0.290)	(0.339)
N	6,706,035	4,090,973
Past Settlements Instrument Only	-0.953***	-0.749*
	(0.267)	(0.330)
N	6,706,035	4,090,973
Excluding Border Areas with Venezuela*	-1.019*	-0.768
	(0.414)	(0.559)
N	6,577,923	4,015,648
Excluding Bogotá	-0.777***	-0.470**
	(0.180)	(0.173)
N	4,338,192	2,619,237
Adjusting ΔM_l Denominator Using 2005 Census	-0.639***	-0.440*
	(0.168)	(0.203)
N	6,706,035	4,090,973
Including Additional Controls$\star$	-0.827***	-0.689*
	(0.175)	(0.326)
N	6,083,175	4,090,973
Using Real Wages		-0.520*
		(0.207)
N		4,090,973
Top-Coding Local Wages Above 99th Percentile		-0.605*
		(0.241)
N		4,090,973

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Reported coefficients are θ_{2018}. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. This Table reports the coefficients of the second-stage regression of the instruments with the immigration shock $\Delta M_{l,2018}$. The outcome is the difference with the base period. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. *The border areas are Cúcuta, Maicao and Arauca. $\star$ Further controls refer to dummies of seven wage quantiles. The sample is restricted to natives aged 25 to 55. Standard errors are clustered at the level of 109 geographic units. Workers are observed in August of each year. Source: PILA, 2015–2018.

Table F.7: Employment and Wage Estimates by Age and Job Tenure

Worker's Age	Below 35 years		Above 35 years	
Job Tenure	0 to 4 years	5 to 9+ years	0 to 4 years	5 to 9+ years
Prob. of Employment	-0.138	0.209	-1.009**	-0.302***
	(0.195)	(0.226)	(0.315)	(0.086)
N	2,099,147	344,156	2,075,913	1,083,435
Wages	-0.479	-0.664*	-0.556	-0.194
	(0.344)	(0.279)	(0.354)	(0.182)
N	1,094,691	240,058	1,170,322	785,839
Clusters	109	109	109	109

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to native employees between 25 and 55 years old. Dependent variables are employment relative to the pre-shock period and wages relative to the base period. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. I constructed pre-shock job tenure from 2007 to 2015. Standard errors are clustered at the level of 109 geographic units. Workers are observed in August of each year. Source: PILA, 2013–2018.

Table F.8: IV Estimates of Department Mobility by Age Group

Age group	25–30	30–35	35–40	40–45	45–50	50–55
Probability of moving	0.127	0.020	-0.093	-0.199	-0.238	-0.261
	(0.381)	(0.376)	(0.329)	(0.286)	(0.247)	(0.195)
N	1,299,106	1,080,584	907,662	761,287	701,270	583,323

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Reported coefficients are θ_{2018}, measuring changes in 2018 relative to 2015. The outcome is an indicator for changing department between 2015 and 2018. Regressions are estimated separately by baseline age group for native formal workers aged 25 to 55 in 2015 who are observed in both years and have nonmissing department codes in both years. Controls include sex and an indicator for employee status in 2015. The immigration shock is instrumented with distance to the nearest border crossing, its square, and past Venezuelan settlements. Standard errors are clustered by baseline FUA. For municipality codes identified as erroneous in 2018, the available February 2020 location is used. Workers are observed in August of each year. Source: PILA, 2015–2018.

Table F.9: Employment and Wage Estimates by Firm Size and Age of Firm

Firm's size	1 to 19 workers		Above 19 workers	
Age of firm	0 to 4 years	5 to 9+ years	0 to 4 years	5 to 9+ years
Prob. of Employment	-0.762**	-0.757***	-1.015**	-0.176
	(0.279)	(0.156)	(0.347)	(0.170)
N	479,715	498,842	923,272	3,700,822
Wages	-1.021*	-0.554	-0.603*	-0.395
	(0.432)	(0.305)	(0.304)	(0.286)
N	274,728	352,015	444,586	2,219,581
Clusters	109	109	109	109

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: I report the θ_{2018} coefficients from Equation (4), estimated separately by subgroups. Wages are relative to 2015, and employment is relative to average employment in 2013–2015. The sample is restricted to native employees between 25 and 55 years old. Dependent variables are employment relative to the pre-shock period and wages relative to the base period. Firm age is measured by the number of years in which the firm is observed in PILA, allowing for gaps. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. Workers are observed in August of each year. Source: PILA, 2013–2018.

Table F.10: Descriptive Statistics for Native Workers by Quintiles of Treatment Effects

(a) Formal Employment

	Q1	Q2	Q3	Q4	Q5
Male share	0.7	0.6	0.5	0.5	0.5
Age of Worker	42.8	40.3	38.5	35.1	31.1
Job Tenure (1-9 years)	2.3	3.6	4.4	4.1	2.8
Monthly Wages (USD)	324.8	462.6	521.8	478.4	336.2
Median Firm Size	79	105	276	510	1109
Quantile of Firm FEs (1-7)	3.8	5.3	6.0	6.3	6.5

(b) Formal Wages

	Q1	Q2	Q3	Q4	Q5
Male share	0.6	0.6	0.6	0.6	0.5
Age of Worker	36.6	38.5	38.8	38.1	37.5
Job Tenure (1-9 years)	3.2	3.9	4.0	3.8	3.5
Monthly Wages (USD)	559.5	466.2	419.3	379.0	393.7
Median Firm Size	86	189	242	309	892
Quantile of Firm FEs (1-7)	5.7	5.8	5.6	5.5	5.5

Note: These tables report the average or median statistics for quintiles of treatment effects (Q1 is the most affected and Q5 is the least affected) in terms of employment and wages, according to the predictions of the trained causal forest using the held-out sample. I constructed pre-shock job tenure from 2007 to 2015. The wages are transformed from Colombian pesos to USD using 2020 exchange rates from the World Bank. Source: PILA, August 2015.

Table F.11: Event Study Estimates on Pre-Treatment Periods of Figure F.8

	Employment			Wages		
	2012	2013	2014	2012	2013	2014
25 to 30 years	-0.312	-0.458	-0.412	0.284	0.370	-0.019
	(0.702)	(0.655)	(0.428)	(0.461)	(0.234)	(0.124)
30 to 35 years	-0.249	-0.512	-0.305	0.104	0.226	-0.011
	(0.499)	(0.415)	(0.322)	(0.166)	(0.235)	(0.272)
35 to 40 years	-0.212	-0.381	-0.060	0.032	0.166	-0.011
	(0.364)	(0.327)	(0.196)	(0.296)	(0.253)	(0.273)
40 to 45 years	0.043	-0.155	-0.328	-0.012	-0.068	-0.501**
	(0.367)	(0.318)	(0.240)	(0.297)	(0.275)	(0.160)
45 to 50 years	0.191	-0.092	-0.101	0.110	0.566	-0.032
	(0.303)	(0.266)	(0.215)	(0.290)	(0.356)	(0.187)
50 to 55 years	0.121	0.005	0.103	0.413	0.369	-0.209
	(0.335)	(0.262)	(0.191)	(0.293)	(0.313)	(0.201)
Males	-0.449	-0.715	-0.473	0.272	0.317	-0.018
	(0.512)	(0.481)	(0.322)	(0.235)	(0.285)	(0.170)
Females	0.297	0.193	0.094	-0.011	0.199	-0.225*
	(0.376)	(0.310)	(0.223)	(0.209)	(0.178)	(0.095)

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: I reduce the sample to a 10% random subsample of the entire dataset due to computational burden. The sample is restricted to natives aged 25 to 55. Controls include interactions of sex with six age categories and a dummy for self-employment in the base period. Standard errors are clustered at the level of 109 geographic units. I observe workers in August of each year. Source: PILA, 2012–2015.

Table F.12: Event Study Estimates on Pre-Treatment Periods of Figure 3

	Employment			Wages		
	2012	2013	2014	2012	2013	2014
Minimum wage	0.313	0.296	0.217	-0.538	-0.409	-0.526
	(0.269)	(0.213)	(0.181)	(0.825)	(0.700)	(0.435)
40th–50th	-0.295	-0.886	-0.190	0.262	0.190	-0.107
	(0.502)	(0.471)	(0.403)	(0.388)	(0.449)	(0.235)
50th–60th	-0.235	-0.766*	-0.284	0.058	0.234	-0.141
	(0.469)	(0.382)	(0.240)	(0.335)	(0.298)	(0.170)
60th–70th	-0.244	-0.100	-0.136	-0.360	0.401*	0.150
	(0.319)	(0.321)	(0.226)	(0.264)	(0.186)	(0.120)
70th–80th	-0.243	-0.475	-0.553**	0.918	0.730*	0.033
	(0.301)	(0.281)	(0.211)	(0.481)	(0.341)	(0.259)
80th–90th	-0.130	-0.432	-0.385*	0.435	0.367	-0.026
	(0.330)	(0.241)	(0.167)	(0.596)	(0.485)	(0.268)
90th–100th	0.330	-0.220	-0.146	-0.039	0.132	-0.477
	(0.483)	(0.173)	(0.136)	(0.288)	(0.269)	(0.297)

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: I reduce the sample to a 10% random subsample of the entire dataset due to computational burden. The sample is restricted to natives aged 25 to 55. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. I observe workers in August of each year. Source: PILA, 2012–2015.

Table F.13: Event Study Estimates on Pre-Treatment Periods of Figure 4

	Employment			Wages		
	2012	2013	2014	2012	2013	2014
1-4 workers	-0.070	-0.022	-0.193	0.263	0.476	0.264
	(0.353)	(0.456)	(0.368)	(0.534)	(0.389)	(0.262)
5-9 workers	-0.044	0.131	-0.484	-0.041	0.176	-0.088
	(0.481)	(0.433)	(0.360)	(0.300)	(0.570)	(0.222)
10-19 workers	-0.314	-0.352	-0.446	0.646	1.156**	0.240
	(0.736)	(0.493)	(0.322)	(0.500)	(0.356)	(0.188)
20-49 workers	-0.525	-0.573	-0.397	0.511*	0.638**	0.398*
	(0.607)	(0.622)	(0.384)	(0.220)	(0.213)	(0.177)
50-99 workers	-0.178	-0.565	-0.497	0.708**	0.877***	0.199
	(0.656)	(0.543)	(0.435)	(0.240)	(0.193)	(0.186)
100 to 999 workers	-0.168	-0.499	-0.211	0.648	0.583	-0.137
	(0.695)	(0.608)	(0.413)	(0.620)	(0.443)	(0.223)
More than 1000 workers	-0.239	-0.462	-0.168	0.202	0.465	0.134
	(0.478)	(0.474)	(0.362)	(0.390)	(0.344)	(0.212)

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: I reduce the sample to a 10% random subsample of the entire dataset due to computational burden. The sample is restricted to native employees between 25 and 55 years old. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. I observe workers in August of each year. Source: PILA, 2012–2015.

Table F.14: Event Study Estimates on Pre-Treatment Periods of Figure 6

	Employment			Wages		
	2012	2013	2014	2012	2013	2014
Lowest quantile	-0.170	-0.169	-0.217	0.121	0.352	-0.867
	(0.469)	(0.483)	(0.426)	(1.819)	(1.369)	(0.750)
2nd quantile	0.040	0.273	0.435	0.390	0.370	0.268
	(0.433)	(0.401)	(0.340)	(0.355)	(0.315)	(0.150)
3rd quantile	-0.695	-0.900	-1.053*	0.574**	0.369	0.049
	(0.548)	(0.537)	(0.457)	(0.201)	(0.230)	(0.120)
4th quantile	0.196	0.455	-0.084	-0.117	0.135	-0.248
	(0.482)	(0.438)	(0.272)	(0.235)	(0.277)	(0.140)
5th quantile	-0.470	-0.772	-0.466	0.480	0.645	0.094
	(0.462)	(0.451)	(0.287)	(0.447)	(0.407)	(0.188)
6th quantile	0.102	-0.385	0.119	0.143	0.383	0.244
	(0.416)	(0.430)	(0.292)	(0.221)	(0.249)	(0.204)
Highest quantile	-0.342	-0.650	-0.456	0.394	0.539	-0.063
	(0.378)	(0.362)	(0.263)	(0.446)	(0.341)	(0.157)

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: I reduce the sample to a 10% random subsample of the entire dataset due to computational burden. The sample is restricted to native employees between 25 and 55 years old. Controls include interactions of sex with six age categories and a dummy for self-employed in the base period. Standard errors are clustered at the level of 109 geographic units. I observe workers in August of each year. Source: PILA, 2012–2015.